

DEPLOYMENT PROTOCOL // OMNIBUS EDITION 2026.1

DEPLOYMENT PROTOCOL

ZERO SUBSCRIPTIONS.UNLIMITED.

MASTER INFRASTRUCTURE MANUAL

BUILD VERSION: 2026.1

STATUS: ACTIVE UNRESTRICTED

PLATFORM: iOS / iPadOS / macOS / Hybrid Cloud

ENGINE: Metal FlashAttention 2.0

COMPILED: May 2026

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BASE SYSTEM
ARCHITECTURE

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MASTER MODEL
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EXECUTIVE SUMMARY – READ FIRST

This manual provides complete operational guidance for deploying Draw Things as a production-grade AI image and video generation environment on Apple devices. It is intended for practitioners who require precise, repeatable, field-tested configuration – not introductory guidance.

FULLY OFFLINE, FREE, NATIVE

Zero-cost AI generation. All local generation runs entirely on-device – no internet, no cloud dependency, no subscription for base functionality.

BROADEST MODEL SUPPORT

SD 1.5, SD 2.x, SDXL, FLUX.1, FLUX.2, Qwen Image, Wan 2.1/2.2, LTX-2.3, HiDream, Hunyuan, ERNIE Image. CivitAI integration for thousands more.

PERFORMANCE LEADER

Metal FlashAttention 2.0 – up to 25% faster than mflux, 94% faster than ggml/gguf for FLUX.1, 163% faster than DiffusionKit for SD3.5 Large.

PROFESSIONAL FEATURE SET

LoRA training (on-device), Textual Inversion, ControlNet, inpainting, outpainting, img2img, video generation (T2V + I2V), infinite canvas, pose editing, API server.

VERSION METADATA

Version: 2026.1 – Platform: iOS / iPadOS / macOS – Status: ACTIVE – Engine: Metal FlashAttention 2.0 – App Version: 1.20260501.0 – Distribution: App Store (Free) – Rating: 4.5/5.0 (710+ ratings)

TEN ACTIONABLE DEPLOYMENT DIRECTIVES

Install **FLUX.1 [dev]** as the primary high-quality image generation model on any device with 16GB+ RAM. Current quality ceiling in the Apple ecosystem.

Use **SDXL (8-bit)** as the best-balance model for mid-range devices (6-8GB RAM). Broad LoRA ecosystem, reasonable RAM footprint.

Always pair samplers to model families: Euler A Trailing (index 10) for FLUX/SD3/video; DPM++ 2M Karras (index 0) for SD 1.5/SDXL. Mismatched samplers are the #1 configuration error.

Begin **LoRA training** with 10-15 high-quality images; 500-1000 steps; unique trigger word. Test checkpoints at intermediate steps before overfitting.

Deploy **negative TI embeddings** (<bad_hands_v5>, <easynegative>) in the negative prompt field for SD 1.5/SDXL anatomy improvement.

Enable Server Offload when generating video on iPhone or older iPad. Configure gRPCServerCLI on a local Mac. Ensure version parity.

Use CausVid LoRA (4-12 steps) for rapid video iteration. Disable Causal Inference for 1.3B variant; enable for 14B variant.

Apply multi-pass strategy for production-quality stills: concept → refinement → detail inpaint → final upscale.

For inpainting, describe only the new content – never describe surrounding image or removal action. Model derives context from unmasked pixels.

Exploit CivitAI integration: tap green play button → select "Draw Things" to import any community model, LoRA, or embedding directly.

I

QUICK REFERENCE

Field-tested presets, matrices, and decision trees for rapid deployment without reading the full manual.

QUICK-START SETTINGS BY USE CASE

READY-TO-DEPLOY PRESETS

USE CASE	MODEL	SAMPLER	STEPS	CFG	RESOLUTION	NEGATIVE
Photorealistic portrait	FLUX.1 [dev]	Euler A Trailing (10)	25	2.5	1024×1024	(minimal or omit)
Anime character	Illustrious SDXL	DPM++ SDE Karras (4)	25	6	1024×1024	worst quality, bad anatomy
Landscape / environment	FLUX.1 [dev]	Euler A Trailing (10)	28	3.0	1024×768	blurry, watermark
Fast concept sketch	SDXL Turbo	Euler A (1)	4	1	512×512	(none)
Logo / graphic design	FLUX.1 [schnell]	Euler A Trailing (10)	4	1	1024×1024	(none)
Vintage photograph	SD 1.5 + film LoRA	DPM++ 2M Karras (0)	25	7.5	512×512	digital art, 3D render, CGI
Text-in-image	Qwen Image 1.0	UniPC Trailing (17)	25	1.5	1024×1024	blurry, low quality, distorted text
Short mobile video	Wan 2.1 1.3B	Euler A Trailing (10)	20	3	480p, 25 frames	(none)
Cinematic video (desktop)	Wan 2.1 14B	Euler A Trailing (10)	30	3	720p, 49 frames	(none)
Image-to-video (desktop)	LTX-2.3	Euler A Trailing (10)	40	3	768×512	(none)

DEVICE CAPABILITY MATRIX

HARDWARE DEPLOYMENT LIMITS

DEVICE	RAM	MAX PRACTICAL MODEL	VIDEO CAPABLE	LORA TRAINING	NOTES
iPhone 12	4GB	SD 1.5 (8-bit)	No	No	Minimum supported device; 8-bit models only; limited to SD 1.5 ecosystem
iPhone 14 Pro	6GB	SDXL (8-bit)	Limited	No	Swap storage assists with marginal overruns; SDXL at 8-bit viable
iPhone 15 Pro	8GB	FLUX.1 (quantized)	Wan 1.3B	Experimental	A17 Pro Neural Engine advantage; 20% MFA 2.0 inference gain; LoRA training may crash
iPhone 16 Pro	8GB	Qwen Image (via swap)	Wan 1.3B	Limited	Advanced swap memory management enables larger models than physical RAM suggests
iPad (8GB RAM)	8GB	SDXL / FLUX quantized	Wan 1.3B	SD 1.5 LoRA	Minimum for LoRA training; larger screen improves canvas workflow
M1 iPad Pro / M1 Mac	16GB	FLUX.1 [dev]	Wan 1.3B	SDXL LoRA	Comfortable FLUX.1 [dev] usage; SDXL LoRA training viable
M2 / M3 Mac (18-36GB)	18-36GB	FLUX.1 [dev] full / SD3.5	Wan 14B / Hunyuan	FLUX LoRA	Up to 20% faster on M3 vs. M2 (MFA 2.0 gains); FLUX LoRA training supported
M4 Pro Mac (24GB+)	24GB+	FLUX.1 [dev] 5-bit optimal	Wan 2.2 5B / Hunyuan	FLUX LoRA	Best inference speed per watt on current silicon; 20% MFA 2.0 gain confirmed
M2 Ultra (192GB)	192GB	All models; no quantization required	All video models	All LoRA types	Maximum performance tier; 25% faster than mflux; 94% faster than gguf on FLUX.1

MODEL-SAMPLER PAIRING CHEAT SHEET

IF YOU READ NOTHING ELSE, READ THIS

MODEL FAMILY	PRIMARY SAMPLER	ALT SAMPLER	CFG SCALE	STEPS
SD 1.5	DPM++ 2M Karras (0)	Euler A (1)	7-8	20-30
SD 2.x	DPM++ 2M Karras (0)	UniPC (3)	7-9	25-35
SDXL	DPM++ SDE Karras (4)	UniPC (3)	5-7	20-30
SDXL Turbo	Euler A (1)	TCD (9)	1-2	1-4
FLUX.1 [dev]	Euler A Trailing (10)	DPM++ 2M Trailing (15)	1-3.5	20-28
FLUX.1 [schnell]	Euler A Trailing (10)	DDIM Trailing (16)	1	4
FLUX.2 (Kontext)	Euler A Trailing (10)	UniPC Trailing (17)	1-3.5	20-28
SD3 / SD3.5	Euler A Trailing (10)	DPM++ 2M Trailing (15)	4-7	20-30
Qwen Image	UniPC Trailing (17)	Euler A Trailing (10)	1-2	20-30
Wan 2.1 / 2.2 / LTX-2.3	Euler A Trailing (10)	DDIM Trailing (16)	1-5	20-40
Hunyuan Video	Euler A Trailing (10)	DPM++ SDE Trailing (11)	1-7	20-50

//CRITICAL RULE FLUX.1, FLUX.2, SD3/SD3.5, and video models require Trailing or AYS samplers. Never use DPM++ 2M Karras or Euler A (non-trailing) with flow-based models. Using a legacy sampler with a flow model produces degraded or unusable output. Draw Things displays "Incompatible" warnings as of v1.20250819.0 – heed them.

CFG SCALE QUICK REFERENCE

GUIDANCE SCALE EFFECTS

CFG RANGE	EFFECT	RISK	USE FOR
1.0	No guidance – pure model imagination	Ignores prompt	FLUX.1 [schnell]; SDXL Turbo distilled
1.5-3.5	Loose guidance – creative, soft adherence	May miss prompt details	FLUX.1 [dev]; FLUX.2; flow models
4-7	Moderate guidance – balanced adherence	Minimal at this range	SDXL; SD3.5; most SD-family models
7-9	Strict guidance – close prompt following	Over-saturation beginning	SD 1.5/2.x; character/anatomy-critical outputs
10-15	Very strict – artifact risk	Halos, oversaturation, color burn	Avoid except for specific style effects
15+	Extreme – highly degraded	Severe artifacts; unusable	Do not use for production output

CALIBRATION WORKFLOW

For an unfamiliar model, begin at CFG 7 with a fixed seed. Generate the same prompt at CFG 4, 7, and 10 to observe the guidance curve. Identify the highest CFG value before artifact onset. For FLUX models, calibrate in the 1.5-3.5 range using the same methodology.

LOCAL VS. CLOUD ROUTING DECISION TREE

INFRASTRUCTURE LAYER COMPARISON

LAYER	DEPLOYMENT TARGET	CAPABILITIES UNLOCKED	DRAWBACKS
DRAW THINGS (LOCAL) Everyday / Unlimited	M-Series Apple Silicon	Full offline execution, direct Civitai integration, Flux.2 + Z Image Turbo + Wan video, advanced img2img/inpainting, LoRA training. Zero privacy leaks.	RAM-constrained on mobile; no batch parallelization; limited to Apple ecosystem
GOOGLE COLAB (CLOUD) Heavy Batches / Stress Testing	Free T4/V100-class GPU	Free high-end GPU hours, ComfyUI asynchronous workflows, larger Flux.2 runs when local hardware needs a boost.	Ephemeral instances; setup overhead; requires tunnel configuration; no offline capability

// ROUTING DECISION TREE

Is the task a single image or small batch (<10 images)? → **ROUTE LOCAL** (Draw Things)

Is the task video generation on iPhone/iPad with <8GB RAM? → **ROUTE LOCAL + SERVER OFFLOAD** (gRPCServerCLI on Mac)

Is the task a large batch (>50 images), high-resolution grid, or stress test? → **ROUTE CLOUD** (Google Colab + ComfyUI)

Is the task FLUX.2 at 2048×2048+ on a device with <32GB RAM? → **ROUTE CLOUD**

Is privacy absolute (no data leaves device)? → **ROUTE LOCAL ONLY**

Is the device an M2 Ultra with 192GB? → **ROUTE LOCAL FOR ALL TASKS**

II

FULL OPERATIONAL GUIDE

Complete technical breakdown of the Draw Things deployment pipeline. Every section includes full reference tables, step-by-step protocols, and field-tested parameters.

SECTION 1 – SYSTEM OVERVIEW & ARCHITECTURE

1.1 PLATFORM IDENTITY

ATTRIBUTE	VALUE
Application	Draw Things: Offline AI Art
Developer	Draw Things, Inc. (primary author: Liu Liu)
Initial Release	2022
Current Version	1.20260501.0 (May 2026)
Distribution	Apple App Store – Free
Rating	4.5 / 5.0 (710+ ratings)
App Size	259.5 MB (base; models downloaded separately)
Age Rating	18+
Supported Platforms	iPhone 12+, iPad (all modern), Mac (Apple Silicon recommended; Intel supported)
Category	Photo & Video

OPERATIONAL NOTE
Draw Things is the only application on iOS/macOS that supports both on-device inference AND on-device fine-tuning (LoRA training) for FLUX.1 [dev] – a 12-billion-parameter model. This distinction separates it from all competing implementations in the Apple ecosystem.

1.2 CORE ARCHITECTURE

Draw Things is built on a custom Apple Metal inference engine designated **Metal FlashAttention 2.0**. This engine implements the FlashAttention attention algorithm – an I/O-aware, memory-efficient method for computing the attention mechanism in transformer and diffusion model architectures – natively in Apple's Metal GPU compute shading language.

- > **Fully offline, on-device inference:** The entire diffusion pipeline – text encoding, denoising, VAE decoding – executes locally on device hardware with no network dependency.
- > **Apple Silicon optimization:** Leverages the Apple Neural Engine (ANE), GPU, and CPU under a unified memory architecture. Workloads are distributed optimally across compute units.
- > **BFloat16 emulation:** For devices lacking native BF16 hardware support, Draw Things implements a BFloat16 emulation layer with carefully tuned rounding rules to maintain inference accuracy on older hardware (iPhone 12, M1-generation devices).

- > **On-demand weight loading:** Model weights are loaded progressively during inference, enabling generation of models that exceed available VRAM through managed memory paging to the unified memory pool.
- > **Runtime code generation:** Metal shader code is generated at runtime for improved compiler compatibility and reduced binary size.
- > **FP16 accuracy hardening:** Precision and register precision are carefully tuned to reduce NaN (Not a Number) errors during FP16 inference – a common failure mode in competing implementations.

1.3 PERFORMANCE BENCHMARKS

DEPLOYMENT PARAMETERS – BENCHMARK CONDITIONS

Benchmarks conducted by Draw Things Engineering (January 2025). M3 Pro tested at 5-bit quantization (Draw Things) vs. 4-bit (mflux, DiffusionKit). M4 Pro tested at 5-bit (Draw Things) vs. 8-bit (mflux, ComfyUI+gguf) vs. 4-bit (DiffusionKit). M2 Ultra tested at 8-bit (Draw Things) vs. no quantization (mflux, DiffusionKit, ComfyUI+PyTorch). Figures represent per-iteration speed unless noted.

PERFORMANCE COMPARISON MATRIX

HARDWARE	MODEL TESTED	DRAW THINGS	VS. MFLUX	VS. GGML/GGUF	VS. DIFFUSIONKIT
M2 Ultra (76 GPU, 192GB)	FLUX.1 [dev]	Fastest	+25% faster	+94% faster	–
M2 Ultra (76 GPU, 192GB)	SD3.5 Large	Fastest	–	–	+163% faster
M3 Pro (18 GPU, 18GB)	FLUX.1	+20% inference gain	Faster (5-bit vs 4-bit)	Significant margin	Faster (5-bit vs 4-bit)
M4 Pro (20 GPU, 24GB)	FLUX.1 + SD3.5	+20% inference gain	Faster (vs 8-bit gguf)	Faster (vs 8-bit gguf)	Faster
A17 Pro (iPhone 15 Pro)	FLUX.1 (quantized)	+20% inference gain	N/A (mobile)	N/A (mobile)	N/A (mobile)
Older devices (pre-M3)	FLUX.1	~+2% gain	Marginal advantage	Still faster	Still faster

1.4 MEMORY & QUANTIZATION STRATEGY

Draw Things implements dynamic quantization to enable large model inference on constrained unified memory. Quantization reduces the numerical precision of model weights, reducing RAM requirements at the cost of minor quality degradation.

QUANTIZATION LEVELS

LEVEL	BIT DEPTH	RELATIVE SIZE	QUALITY IMPACT	RECOMMENDED USE CASE
BF16 / FP16 (Full)	16-bit	100% (baseline)	None (reference quality)	M2 Ultra / maximum RAM systems
8-bit (INT8)	8-bit	~50%	Negligible to minimal	Standard desktop Mac; recommended default
6-bit	6-bit	~37%	Very low	Mid-range iPad / Mac with constrained RAM
5-bit	5-bit	~31%	Low	M3/M4 Pro with 18-24GB; best balance for FLUX.1
4-bit (INT4)	4-bit	~25%	Moderate; some detail loss	iPhone 15 Pro / 6-8GB devices for SDXL
BF16 Emulation	16-bit emulated	100%	Slight rounding deviation	Older devices lacking native BF16 hardware

FIELD TIP – QUANTIZATION SELECTION

For FLUX.1 on M3/M4 devices, 5-bit quantization is the recommended operating point – it delivers the best balance of generation quality, RAM utilization, and inference speed. 8-bit is the recommended default for M2 and M1 desktop systems. Reserve 4-bit for iPhone deployments only.

1.5 CLOUD COMPUTE INTEGRATION

Draw Things offers an optional subscription tier – **Draw Things+** – providing access to cloud-based generation when local hardware is insufficient for desired model or resolution targets.

- > **Privacy Pass (v1.20250819.0):** Implements an unlinkable anonymization protocol for cloud generation requests – preventing linkage of generations to specific user identities. Zero-knowledge attestation architecture.
- > **Bridge Mode (v1.20251014.0):** Enables the in-app API server to bridge external tool integrations, enabling automation and workflow pipelines via local API endpoints.
- > **Boost Credits:** Allow Draw Things+ subscribers to exceed standard Cloud Compute task limits for burst workloads.
- > **Device Attestation:** Verifies that cloud requests originate from a legitimate Draw Things installation on an Apple device. Currently tags requests for quality assurance; does not restrict access.
- > **gRPC Server CLI:** A companion command-line tool for Mac that enables a local device to act as a compute server for iOS clients (Server Offload Mode). Updated in lockstep with app versions.

SECTION 2 – MODEL ARCHITECTURE & ROUTING

2.1 MODEL FAMILY OVERVIEW

OPERATIONAL NOTE

RAM requirements listed are minimum for generation. Active use RAM may be 10-20% higher due to VAE, text encoder, and system overhead. Sampler and CFG recommendations represent operational defaults – see Section 4 for full sampler reference.

COMPLETE MODEL REFERENCE

MODEL NAME	ARCHITECTURE	PARAMS	IDEAL RES	STEPS	CFG	SAMPLER	RAM REQ	NOTES
Stable Diffusion 1.5	UNet	860M	512×512	20–30	7–8	DPM++ 2M Karras / Euler A	2GB+	Original baseline; widest LoRA/TI compatibility; massive CivitAI ecosystem
Stable Diffusion 2.1	UNet	900M	768×768	25–35	7–9	DPM++ 2M Karras	3GB+	Better coherence than 1.5; reduced LoRA ecosystem; OpenCLIP encoder
SDXL Base	UNet	3.5B	1024×1024	20–30	5–7	DPM++ SDE Karras / UniPC	8GB+	High quality; supports refiner pipeline; dual CLIP text encoder
SDXL Turbo	UNet (distilled)	3.5B	512–1024px	1–4	1–2	Euler A	8GB+	Speed-optimized adversarial distillation; single-step viable
Pony Diffusion / Illustrrious	SDXL finetune	3.5B	1024×1024	20–30	5–7	DPM++ SDE Karras	8GB+	Anime/illustration specialist; score-based tagging system; large community LoRA library
FLUX.1 [dev]	DiT hybrid (12B)	12B	1024×1024+	20–28	1–3.5	Euler A Trailing	16GB+	Open-weight; best available quality; non-commercial license; on-device training supported
FLUX.1 [schnell]	DiT (distilled)	12B	1024×1024+	1–4	1	Euler A Trailing	16GB+	Apache 2.0 license; fastest FLUX variant; commercial use permitted

MODEL NAME	ARCHITECTURE	PARAMS	IDEAL RES	STEPS	CFG	SAMPLER	RAM REQ	NOTES
FLUX.2 (Kontext)	DiT hybrid	12B	1024×1024+	20-28	1-3.5	Euler A Trailing	16GB+	Image editing specialist; in-context editing capability
SD3 Medium / SD3.5 Large	DiT (rectified flow)	2-8B	1024×1024	20-30	4-7	Euler A Trailing	8-16GB	Rectified flow architecture; excellent text rendering; trimodal CLIP
Qwen Image 1.0	Transformer (LLM-based)	72B-equiv.	Variable	20-30	1-2	UniPC Trailing	20GB+	Advanced text rendering; multi-image referencing; 4-step and 8-step acceleration LoRAs
Wan 2.1 / 2.2 (1.3B)	DiT video	1.3B	480p	20-40	1-5	Euler A Trailing	8GB+	Lightweight video; T2V and I2V; iPhone-viable; CausVid LoRA for 4-12 step generation
Wan 2.1 (14B)	DiT video	14B	720p	20-40	1-5	Euler A Trailing	32GB+	High-quality video; 81 frames max; requires desktop Mac; CausVid LoRA supported
LTX-2.3	DiT video (2B)	2B	768×512	40	3	Euler A Trailing	8GB+	Real-time-capable; sound support; T2V and I2V; 257+ frames
HiDream	Transformer	Variable	1024×1024	20-30	4-7	Euler A Trailing	12GB+	High quality; DiT-based; newer architecture; strong community adoption
Kwai Kolors	SDXL-based	3.5B	1024×1024	20-30	5-7	DPM++ SDE Karras	8GB+	Excellent portraits and realistic humans; inpainting variant available
Hunyuan Video	DiT video	Variable	720p	20-50	1-5	Euler A Trailing	16GB+	Strong motion coherence; 1+4n frame constraint; T2V + I2V

MODEL NAME	ARCHITECTURE	PARAMS	IDEAL RES	STEPS	CFG	SAMPLER	RAM REQ	NOTES
ERNIE Image	Transformer	Variable	Variable	20-30	Variable	Euler A Trailing	Variable	Latest addition (BAIDU, 2026); T2I and I2V; newest pipeline in active deployment

2.2 MODEL SELECTION PROTOCOL

Use the following decision hierarchy to select the appropriate model for a given task and hardware configuration:

Step 1 – Determine Available RAM. Check device RAM (see Section 12.2). RAM is the primary constraint. Less than 4GB: SD 1.5 (8-bit) only. 6-8GB: SDXL (8-bit) or FLUX (quantized). 16GB+: FLUX.1 full or SD3.5. 32GB+: Wan 14B video.

Step 2 – Determine Task Type. Still image: proceed to Step 3. Video generation: select from Section 10 video models. Image editing/inpainting: FLUX.2 / FLUX.1 Fill preferred.

Step 3 – Determine Quality vs. Speed Priority. Maximum quality (16GB+ RAM): FLUX.1 [dev]. Maximum speed (any device): SDXL Turbo (4 steps) or FLUX.1 [schnell] (4 steps). Balanced (8GB+ RAM): SDXL (8-bit) or FLUX.1 (quantized).

Step 4 – Determine Style Domain. Photorealistic: FLUX.1 [dev] or SD 1.5+realistic LoRA. Anime/illustration: Illustrrious XL or Pony Diffusion. Portrait: Kwai Kolors or FLUX.1 [dev]. Text-in-image: Qwen Image or FLUX.1. Video: Wan 2.1 1.3B (mobile) or Wan 2.1 14B / Hunyuan (desktop).

Step 5 – Confirm LoRA Compatibility. SD 1.5 LoRAs: SD 1.5 only. SDXL LoRAs: SDXL/Pony/Illustrrious. FLUX LoRAs: FLUX.1/FLUX.2 only. Acceleration LoRAs: model-specific (see Section 5.6).

2.3 MODEL INSTALLATION PROCEDURE

Method A – In-App Library: Open Draw Things → tap the model name at top of generation panel → tap "Model Manager" → browse or search available models → tap download icon. Model downloads directly to device storage.

Method B – CivitAI Import: Browse CivitAI.com in Safari or the CivitAI app → locate desired model or LoRA → tap the green play/download button on the model card → select "Draw Things" from the app selector → confirm import. Draw Things opens and begins downloading automatically.

Method C – Local .safetensors Import: Obtain .safetensors model file via Files app or AirDrop → in Draw Things, tap Model Manager → tap the import icon (top right) → navigate to file → select. Draw Things will index and register the model.

Conversion to 8-bit: In Model Manager, long-press (or right-click on Mac) an imported full-precision model → select "Convert to 8-bit" → confirm. Conversion runs on-device. Original file preserved unless manually deleted.

//WARNING – STORAGE REQUIREMENTS

FLUX.1 [dev] at 8-bit requires approximately 12-14GB of device storage. SD3.5 Large at 8-bit requires approximately 8GB. Ensure sufficient storage before initiating large model downloads. Models are stored in the Draw Things app container and are not visible in Files by default.

2.4 MODEL ROUTING LOGIC

- > **SDXL Base + Refiner:** Use the Base model for the full generation pass (20-30 steps). Route the output to the SDXL Refiner at approximately 80% image noise strength for detail enhancement (10-15 additional steps). Do not use the Refiner alone for cold generation.
- > **8-bit vs. Full Precision:** Default to 8-bit on any system with less than 48GB RAM for FLUX-class models. Full precision recommended only on M2 Ultra (192GB) or similar where quality differences are production-critical.
- > **Sampler Compatibility Warnings:** As of version 1.20250819.0, Draw Things displays "Incompatible" warnings when a selected sampler is inappropriate for the active model family. Heed these warnings – using a legacy sampler (e.g., PLMS) with a flow model (FLUX, SD3) produces degraded or unusable output.
- > **Newer Models vs. Older Samplers:** FLUX.1, FLUX.2, SD3/SD3.5, and video models require Trailing or AYS samplers. Never use DPM++ 2M Karras or Euler A (non-trailing) with flow-based models. Use Euler A Trailing (index 10) as the universal default for all flow-matching architectures.

SECTION 3 — PROMPT ARCHITECTURE

3.1 PROMPT SYSTEM FUNDAMENTALS

The text encoder translates human-language prompts into numerical embeddings that guide the diffusion denoising process. Different model families use fundamentally different text encoders, which determines optimal prompt structure and vocabulary:

TEXT ENCODER COMPARISON

MODEL FAMILY	TEXT ENCODER	TOKEN LIMIT	PROMPT STYLE
SD 1.5 / 2.x	CLIP ViT-L/14	77 tokens	Comma-separated keywords; quality tags effective; strict token limit
SDXL	Dual CLIP (ViT-L + ViT-bigG)	77+77 tokens	Natural language OR keywords; quality tags unnecessary; richer semantic understanding
FLUX.1 / FLUX.2	T5-XXL + CLIP-L	512+ tokens	Long-form natural language; complex spatial relationships understood; exceptional adherence
SD3 / SD3.5	T5-XXL + CLIP-L + CLIP-G	512+ tokens	Natural language; excellent text rendering; multi-subject support
Qwen Image	LLM-style tokenizer	Extended	Descriptive sentences; LLM-style instructions; multi-image referencing supported

3.2 PROMPT CONSTRUCTION BY MODEL FAMILY

SD 1.5 / 2.X — KEYWORD PROTOCOL

Comma-separated keywords are the preferred format. The CLIP encoder responds more reliably to dense keyword lists than to natural language sentences. Quality booster tags significantly impact output in this architecture.

Recommended structure: [Subject] + [Style/Medium] + [Quality Tags] + [Lighting] + [Composition]

portrait of a woman, oil painting, masterpiece, best quality, highly detailed, soft studio lighting, close-up, bokeh background

Effective quality boosters: masterpiece, best quality, highly detailed, 8k, sharp focus, professional photography

SDXL — NATURAL LANGUAGE PROTOCOL

SDXL's dual CLIP encoders provide significantly richer semantic understanding. Quality booster tags are unnecessary and may produce redundant emphasis. Focus on descriptive, specific language.

A futuristic cityscape at dusk, neon reflections on wet pavement, cinematic wide shot, dramatic atmospheric perspective, warm amber light from storefronts contrasting with blue twilight sky

Prompt weighting ((word:1.5)) has reduced sensitivity in SDXL compared to SD 1.5. Use sparingly.

FLUX.1 / FLUX.2 – LONG-FORM NATURAL LANGUAGE PROTOCOL

FLUX's T5-XXL text encoder understands complex spatial relationships, object attributes, and multi-clause descriptions with exceptional fidelity. Long, detailed prompts are not only acceptable but recommended.

A weathered lighthouse standing on a rocky coastal cliff at sunset, waves crashing thirty feet below against jagged granite formations, golden hour light casting long shadows eastward across tide pools, a small fishing vessel visible on the horizon, cinematic photography style, highly detailed

Quality booster tags have minimal effect and are generally unnecessary. Negative prompts have significantly reduced effectiveness due to guidance distillation.

QWEN IMAGE – LLM-STYLE DESCRIPTIVE PROTOCOL

Qwen Image uses an LLM-derived tokenizer, enabling instruction-style and descriptive natural language prompting. Particularly powerful for text-in-image generation and complex compositional scenes.

Generate an image of a modern coffee shop interior with warm lighting. On a chalkboard menu behind the counter, write the text "SPECIALTY COFFEE – Single Origin Espresso \$4.50" in chalk lettering. Include a barista in the foreground.

3.3 PROMPT WEIGHTING & ATTENTION

Draw Things supports prompt weighting syntax for SD-family models using parenthetical notation:

- > (word:1.5) – Increases attention weight by 50% (emphasizes)
- > (word:0.5) – Decreases attention weight by 50% (de-emphasizes)
- > [word] – Mild de-emphasis (some implementations)
- > Nesting: ((word:1.2):1.3) – Multiplied weight (use cautiously; can destabilize)

//WARNING – FLUX WEIGHTING INCOMPATIBILITY

FLUX.1, FLUX.2, and FLUX-derived models do NOT support traditional parenthetical weighting syntax. The T5-XXL encoder processes tokens differently; weighting syntax will appear as literal characters or produce undefined behavior. For FLUX models, rephrase and reposition descriptors instead of using weight modifiers.

3.4 NEGATIVE PROMPT ARCHITECTURE

Negative prompts leverage the Classifier-Free Guidance (CFG) mechanism: the model denoises simultaneously in a "positive" direction (toward the prompt) and a "negative" direction (away from the negative prompt), then extrapolates between them at the CFG scale.

NEGATIVE PROMPT BY MODEL FAMILY

MODEL FAMILY	RECOMMENDED NEGATIVE PROMPT	EFFECT LEVEL
SD 1.5	ugly, deformed, blurry, low quality, worst quality, bad anatomy, extra limbs, duplicate, watermark, text, signature, cropped, out of frame	High – strongly impacts output quality
SD 2.x	ugly, deformed, blurry, low quality, watermark, text, extra limbs, bad anatomy	High
SDXL	blurry, low quality, watermark, deformed	Medium – simpler negatives sufficient
SDXL Turbo	(generally omit – distilled guidance reduces effectiveness)	Low
Pony / Illustrious	worst quality, bad anatomy, censored, jpeg artifacts, low resolution, blurry, deformed hands	High – score-based tags important
FLUX.1 / FLUX.2	(omit entirely or use minimal: blurry, low quality)	Very low – guidance distillation negates negative prompts
SD3 / SD3.5	blurry, low quality, watermark, deformed	Medium
Qwen Image	blurry, low quality, distorted text, watermark	Medium
Video Models (Wan/LTX/Hunyuan)	static, low quality, blurry motion, flickering, watermark	Low-Medium

FIELD NOTE – NEGATIVE EMBEDDINGS

For SD 1.5/SDXL models, placing Textual Inversion (TI) embedding files in the negative prompt field – such as <bad_hands_v5> or <easynegative> – provides significantly more targeted quality improvement than keyword negatives alone. See Section 6.5 for full negative embedding protocol.

3.5 SEED MANAGEMENT

- > **Fixed Seed:** Manually set a specific seed value in the generation panel. Use for iterative prompt refinement without changing composition.
- > **Random Seed (-1):** Default. New random seed each generation. Use for exploration.
- > **Scale Alike Mode:** Draw Things' proprietary seed mode. Maintains consistent composition when changing image resolution – the generated composition scales proportionally rather than resampling to a different arrangement. Essential for resolution testing workflows.

> **AUTOMATIC1111 Compatible Seed Mode:** For cross-platform reproducibility, Draw Things offers an A1111-compatible seed mode that produces identical results to AUTOMATIC1111 web UI for the same seed/model combination on SD 1.5 models.

SECTION 4 – SAMPLER CONFIGURATION MATRIX

4.1 SAMPLER THEORY

Diffusion models generate images by reversing a noise-addition process. Starting from a tensor of pure random noise, the model iteratively predicts and removes noise across a defined number of timesteps – called the *denoising schedule*. The **sampler** (also called solver) determines the mathematical algorithm used for each denoising step.

- > **ODE Solvers (Euler, Heun, PLMS):** Treat the denoising as a deterministic ordinary differential equation. Same seed produces identical output every run. Efficient; convergent. Euler is first-order; Heun is second-order (more accurate but 2× cost per step).
- > **Ancestral Samplers (Euler A, DPM++ SDE):** Add stochastic noise at each step. Do not converge – output varies even with fixed seed as step count changes. Produces more varied, creative results. Generally preferred for artistic exploration.
- > **Karras Noise Schedule:** A variance-preserving noise schedule developed by Karras et al. that distributes denoising steps more efficiently across the noise range, enabling higher quality at lower step counts.
- > **Trailing Schedule (Flow Models):** FLUX, SD3, and video DiT models use rectified flow matching rather than DDPM diffusion. Trailing samplers are modified to match the timestep spacing requirements of flow-matching architectures. *Do not use non-trailing samplers with flow models.*
- > **AYS (Align Your Steps):** Adaptive step scheduling that concentrates denoising compute at the most impactful timesteps. Enables high quality at reduced step counts (10-20 steps vs. 25-30 standard).

4.2 COMPLETE SAMPLER REFERENCE TABLE

ALL 19 SAMPLERS

INDEX	NAME	TYPE	BEST FOR	STEPS	NOTES
0	DPM++ 2M Karras	Multi-step adaptive	SD 1.5 / SDXL general purpose	20-30	Default recommendation for SD-family; convergent; consistent results
1	Euler A	Ancestral ODE	Creative exploration; fast iteration; SD 1.5	20-30	Does not converge; results vary with step count even at fixed seed; high creativity variance
2	DDIM	Deterministic	Reproducible results; low-step generation	10-20	Original SD sampler; fully deterministic; inpainting-compatible; low compute cost
3	UniPC	Predictor-corrector	Portraits; consistency; SDXL	15-25	Efficient multi-step; high quality at moderate steps; good for structured compositions

INDEX	NAME	TYPE	BEST FOR	STEPS	NOTES
4	DPM++ SDE Karras	Stochastic	High detail; realism; SD 1.5/SDXL	20-30	Slower than DPM++ 2M; most detailed output for SD-family; ancestral noise adds texture richness
5	PLMS	Linear multi-step	Simple tasks; legacy compatibility	20-30	Outdated; superseded by DPM++ family; retained for legacy model compatibility only
6	LCM	Consistency model	LCM-specific models and LoRAs only	1-4	Ultra-fast; requires LCM-distilled model or LCM LoRA; incompatible with standard models
7	Euler A Substep	Enhanced ancestral	Complex scenes; SD 1.5/SDXL detail	20-30	Sub-step refinement adds detail vs. standard Euler A; moderate speed penalty
8	DPM++ SDE Substep	Enhanced stochastic	Refined high-quality outputs; SD 1.5/SDXL	20-30	Quality ceiling higher than standard DPM++ SDE; slowest SD sampler; best results
9	TCD	Distilled	Few-step generation; TCD-distilled models	4-8	Trajectory Consistency Distillation; fast alternative to LCM; slightly more flexible step range
10	Euler A Trailing	Flow-model ancestral	FLUX.1/FLUX.2; SD3/SD3.5; Qwen; video	20-28	Universal flow-model default. Trailing timestep schedule matches rectified flow architecture; do not substitute non-trailing samplers
11	DPM++ SDE Trailing	Stochastic flow	Video/flow models; motion variation	20-28	Stochastic variation useful for video motion diversity; adds temporal noise variation
12	DPM++ 2M AYS	Adaptive multi-step	Consistent quality; reduced steps	10-20	AYS step alignment enables 15-step outputs matching 30-step DPM++ 2M quality; efficiency gain
13	Euler A AYS	Adaptive ancestral	Creative + stable; SD 1.5 quality at lower steps	15-25	Balanced creativity and convergence; good for portrait and character generation
14	DPM++ SDE AYS	Adaptive stochastic	Quality + diversity; SD 1.5/SDXL	15-25	Best output diversity at reduced step counts; useful for batch exploration runs
15	DPM++ 2M Trailing	Trailing multi-step	High-resolution flow model outputs	20-30	Efficient high-res generation with flow models; alternative to Euler A Trailing for structured outputs
16	DDIM Trailing	Deterministic trailing	Video frame consistency; flow model reproducibility	10-20	Deterministic; excellent temporal coherence for video; fast flow-model inference
17	UniPC Trailing	Trailing predictor	Detail preservation; Qwen Image; flow models	15-25	Consistent detail and structure in flow model outputs; recommended for Qwen Image

CFG RANGE	EFFECT	RISK	RECOMMENDED USE
15+	Extreme – highly degraded	Severe artifacts; unusable	Do not use for production output

FIELD TIP – CFG CALIBRATION WORKFLOW

For an unfamiliar model, begin calibration at CFG 7 with a fixed seed. Generate the same prompt at CFG 4, 7, and 10 to observe the guidance curve. Identify the highest CFG value before artifact onset. For FLUX models, calibrate in the 1.5-3.5 range using the same methodology.

SECTION 5 – LORA INTEGRATION PROTOCOL

5.1 WHAT IS LORA

Low-Rank Adaptation (LoRA) is a parameter-efficient fine-tuning technique that injects small, trainable weight matrices (adapters) into specific layers of a pretrained model. During inference, these adapter weights are merged with the base model weights – altering the model's output style, subject knowledge, or concept understanding – without modifying the underlying base model weights on disk.

Key properties: small file size (typically 10-200MB vs. multi-GB base models), stackable (multiple LoRAs can be active simultaneously), and reusable across different generation sessions. LoRAs do not require retraining the base model; they function as lightweight "personality overlays."

5.2 LORA ARCHITECTURE COMPATIBILITY TABLE

LORA TYPE COMPATIBILITY

LORA TYPE	COMPATIBLE BASE MODELS	TRIGGER WORD REQUIRED	TYPICAL WEIGHT RANGE	NOTES
SD 1.5 LoRA	SD 1.5 only	Yes (usually)	0.5-1.0	Widest selection on CivitAI; massive community library; many styles and characters
SDXL LoRA	SDXL / Pony / Illustrious	Yes (usually)	0.5-1.0	Higher RAM training requirement; growing library; supports SDXL ecosystem broadly
FLUX LoRA	FLUX.1 [dev] / FLUX.2	Sometimes	0.8-1.2	Newer architecture; on-device training supported in Draw Things; growing ecosystem
Acceleration LoRA (LCM)	LCM-compatible models	No	1.0	Reduces generation to 1-4 steps; requires LCM sampler (index 6)
Acceleration LoRA (TCD)	TCD-compatible SD/SDXL	No	1.0	4-8 step generation; more flexible than LCM; use TCD sampler (index 9)
CausVid LoRA	Wan 2.1 (1.3B and 14B)	No	1.0	Enables 4-12 step video generation; causal temporal compression; model-variant dependent settings
Inpaint LoRA	SDXL base (Fooocus Inpaint)	Sometimes	0.6-1.0	Style application during inpainting pass; pairs with inpaint-capable base
Qwen Acceleration LoRA	Qwen Image 1.0	No	1.0	4-step and 8-step variants available; maintains output quality at reduced steps

5.3 LORA INSTALLATION PROCEDURE

Step 1 – Source Selection: From in-app library: Model Manager → LoRA tab → browse/download. From CivitAI: navigate to model page → tap green play button → select "Draw Things" → confirm in-app download. From local file: Files app → locate .safetensors → share to Draw Things OR Model Manager → import icon.

Step 2 – Trigger Word Configuration: After import, Draw Things prompts for trigger word entry. Enter the LoRA's specified trigger word exactly (check CivitAI model page). Trigger word is stored with the LoRA entry and auto-inserted when the LoRA is activated.

Step 3 – Weight Setting: In the generation panel, locate the LoRA section → set weight slider. Default starting point: 1.0. For style LoRAs: begin at 0.75. For character LoRAs: begin at 0.8–1.0. For acceleration LoRAs: use exactly 1.0.

Step 4 – Multiple LoRA Stacking: Tap the + button in the LoRA section to add additional LoRAs. Each has an independent weight slider. Monitor total combined weight – exceeding 2.0 combined weight across multiple LoRAs risks output destabilization.

Step 5 – Inference Stage Selection (v1.20250807.0+): For SDXL base+refiner pipelines, each LoRA can be assigned to apply during the Base stage, Refiner stage, or All stages. Set via the LoRA entry's stage selector.

5.4 LORA TRAINING PROTOCOL (ON-DEVICE)

DEPLOYMENT PARAMETERS – ON-DEVICE LORA TRAINING

Draw Things is the only iOS/macOS application supporting efficient on-device FLUX.1 LoRA training. Training is available via the PEFT (Parameter-Efficient Fine-Tuning) tab in the app.

ON-DEVICE TRAINING PARAMETERS

PARAMETER	SD 1.5 LORA	SDXL LORA	FLUX.1 LORA
Minimum RAM	8GB (iPad)	16GB (Mac recommended)	16GB (Mac recommended)
Network Dimension	≤16 (default safe)	16–32	16 (low RAM: 8)
Training Steps	500–1000 (beginners)	500–1500	1500–2000 (or fewer; test early)
Dataset Size	10–15 images	10–20 images	5–20 images (multi-res supported)
Resolution	512×512	768×768 or 1024×1024	512×512 to 768×768 (multi-res)
Learning Rate	Default (safe); adjust carefully	Default recommended	0.0004 recommended

PARAMETER	SD 1.5 LORA	SDXL LORA	FLUX.1 LORA
iPhone 15 Pro Viability	Experimental (may crash)	Not recommended	Not recommended

Training Best Practices:

- > Dataset quality supersedes quantity: 10-15 sharp, well-lit, varied-pose images outperform 50 inconsistent images
- > Generate captions using Blip2 or Moondream2 (available in Draw Things) for FLUX LoRA training
- > Trigger word must be unique – avoid celebrity names, common words, or existing model tokens
- > Test checkpoints at intermediate step counts to identify the optimal stopping point before overfitting
- > Overfitting indicators: output becomes "burned" (artifacts, oversaturation, loss of diversity) or highly repetitive regardless of prompt variation
- > For FLUX LoRA, target layers 26, 31, 35, 39 for best likeness training results
- > Enable Metal FlashAttention v2 during training for 20-25% RAM savings

5.5 LORA STRENGTH CALIBRATION

WEIGHT RANGE EFFECTS

WEIGHT RANGE	EFFECT	RECOMMENDED USE
0.3-0.5	Very subtle influence; barely perceptible style shift	Subtle style blending; multi-LoRA stacks
0.5-0.75	Moderate influence; style clearly present; prompt maintains control	Style LoRAs; mood modifiers; multi-stack configurations
0.8-1.0	Standard influence; LoRA concept dominant but prompt-steerable	Character LoRAs; concept LoRAs; standard deployment
1.0-1.2	Strong influence; LoRA dominant; prompt may lose some control	Strong style imposition; single-LoRA configurations
>1.2	Risk of LoRA overpowering – artifacts, color distortion, style bleeding	Avoid; experimental only

5.6 ACCELERATION LORAS

- > **LCM LoRA:** Latent Consistency Model LoRA. Apply to SD 1.5 or SDXL base model. Set sampler to LCM (index 6). Use 1-4 steps. CFG = 1-2. Dramatically reduces generation time; quality lower than full-step generation but useful for concept exploration.
- > **TCD LoRA:** Trajectory Consistency Distillation. Apply to compatible SD/SDXL model. Set sampler to TCD (index 9). Use 4-8 steps. More flexible step range than LCM.

- > **CausVid LoRA for Wan 2.1 1.3B:** Disable Causal Inference in Settings before use. Use 4-12 steps. Even 1-4 steps viable for some prompts. Dramatically reduces video generation time on iPhone/iPad.
- > **CausVid LoRA for Wan 2.1 14B:** Enable Causal Inference in Settings. Better complex motion handling. Smoother realism at equivalent step counts. Available via in-app library or Hugging Face.
- > **Qwen Image 4-step / 8-step Acceleration LoRAs:** Reduces Qwen Image generation from 20-30 steps to 4 or 8 steps respectively. Quality maintained relative to step count reduction. Import via in-app LoRA library.
- > **FLUX Acceleration LoRAs:** FLUX.1 Turbo Alpha, Hyper FLUX.1 [dev] 8-step, FLUX.1 [dev] to [schnell] 4-step conversion. Enable high-quality FLUX outputs at 4-8 steps rather than 20-28.

SECTION 6 – TEXTUAL INVERSION (TI / EMBEDDINGS)

6.1 WHAT IS TEXTUAL INVERSION

Textual Inversion (TI) is a technique that trains a new token – a "word" – in the text encoder's embedding space to represent a specific concept, style, or object. Unlike LoRA, which modifies model layer weights, TI only adds new token embeddings to the text encoder vocabulary. The base model weights remain completely unmodified.

The new token is activated by using its designated name in the prompt (syntax: <embedding_name>). The model then processes this token as it would any trained vocabulary word – but with semantics derived from the training images rather than from natural language.

6.2 TI VS. LORA COMPARISON

TI VS LORA

ATTRIBUTE	TEXTUAL INVERSION (TI)	LORA
File size	~10-50KB	~10-200MB
Training complexity	Lower (text encoder only)	Medium (UNet/transformer layers)
Effect strength	Moderate	Strong
Stacking	Multiple TIs combinable	Multiple LoRAs stackable
Syntax activation	<embedding_name> in prompt	Trigger word in prompt
Best for	Style, mood, concept modifier, quality enhancer	Subject, character, strong style
RAM impact during inference	Minimal (embedding lookup only)	Moderate (layer weight modification)
Base model modification	None	None (adapters only)
Draw Things support since	January 2023 update	Concurrent release

6.3 TI INSTALLATION & ACTIVATION

Step 1 – Import: Via CivitAI: navigate to embedding page → green play button → "Draw Things." Via local file: share .pt or .safetensors embedding file to Draw Things. Draw Things registers it under Embeddings in the model library.

Step 2 – Activate in Prompt: Type <embedding_name> exactly as named in the positive or negative prompt field. Capitalization and spelling must match the filename exactly.

Step 3 – Combine: Multiple TIs can coexist in a single prompt: portrait of a woman <style_impressionist> <lighting_golden_hour> soft focus. Each embedding contributes its trained semantic independently.

6.4 TI USE CASES

- > **Style Modifiers:** Embeddings trained on specific photographic or artistic styles – e.g., analog film grain texture, oil painting impasto technique, watercolor wash
- > **Quality Enhancers:** Quality-boosting embeddings trained to push models toward high-detail, well-composed outputs – particularly effective for SD 1.5 where they function similarly to quality booster keyword tags but more precisely
- > **Character Consistency:** Lightweight character embeddings for repeated reference to a specific character design without full LoRA training investment
- > **Mood Operators:** Embeddings encoding specific emotional or atmospheric qualities – cinematic tension, nostalgic warmth, surreal dreamlike quality

6.5 NEGATIVE EMBEDDINGS

One of the most effective TI applications is the negative embedding: a TI embedding placed in the *negative prompt field* rather than the positive prompt. Negative embeddings encode undesirable features – poor anatomy, low quality artifacts, deformed anatomy – and their placement in the negative field instructs the CFG mechanism to actively steer away from those learned qualities.

RECOMMENDED NEGATIVE EMBEDDINGS FOR SD 1.5

<bad_hands_v5> – Dramatically improves hand anatomy generation. Place in negative prompt field.

<easynegative> – General quality negative; suppresses common low-quality features.

<ng_deepnegative_v1_75t> – Deep negative for anatomy and quality improvement.

Usage: In negative prompt field, type: ugly, blurry <bad_hands_v5> <easynegative>

SECTION 7 — IMAGE-TO-IMAGE (IMG2IMG) PIPELINE

7.1 IMG2IMG FUNDAMENTALS

In standard text-to-image (txt2img) generation, the diffusion process begins from a tensor of pure random noise. In img2img mode, a reference image is first encoded into the latent space by the VAE encoder, noise is added to a specified level (controlled by the Strength parameter), and denoising proceeds from that partially-noised state. The result is a generated image that incorporates both the prompt guidance and the structural information preserved from the original image.

Higher Strength = more noise added initially = less of the original image preserved = closer to txt2img behavior. Lower Strength = less noise = more of original image structure preserved = subtler transformation.

7.2 STRENGTH PARAMETER PROTOCOL

STRENGTH CALIBRATION

STRENGTH %	EFFECT ON INPUT	USE CASE
10-30%	Subtle texture/color shift; composition fully preserved	Style transfer; light color grading; texture adjustment
40-60%	Moderate reinterpretation; major structures retained; details regenerated	Concept transformation; major style change with composition retention
70-85%	Major structural change; only faint composition guidance remains	Complete restyling; near full-regeneration with composition hint
90-100%	Near txt2img behavior; only faint composition ghost from input	Use when only very rough composition seed is desired from original

7.3 IMG2IMG WORKFLOW

Step 1: Upload or load reference image to the Canvas in Draw Things (tap the image import icon or drag from Photos).
Step 2: Select appropriate model matching desired output style. The model and input image style should be compatible (e.g., do not use an anime model on a photorealistic input unless intentional style transfer is the goal).
Step 3: Write prompt describing the desired output — not the input. Include style descriptors, subject description, and any changes from the original.
Step 4: Set Strength. Starting recommendation: 50-70% for balanced edits. Use lower (25-40%) for subtle modifications.

Step 5: Generate. Evaluate result. Adjust Strength up (more change) or down (less change) based on outcome.

Step 6: Apply additional passes progressively. Each pass of img2img at 20-40% strength adds refinement without major structural disruption.

7.4 RESOLUTION HANDLING

img2img respects the source image resolution by default. The diffusion process runs at the input image's dimensions. Changing the output resolution without matching the input can cause composition distortion.

Upscale Workflow:

Generate initial image at native optimal resolution (e.g., 512×512 for SD 1.5, 1024×1024 for SDXL)

Export the generated image

Re-import as img2img input at 2× target resolution (e.g., 1024×1024 or 2048×2048)

Set Strength to 30-50%; write detailed, specific prompt

Generate: the diffusion process fills in high-frequency detail appropriate to the larger canvas size

7.5 INSTRUCT PIX2PIX MODE

The "Edit (Instruct Pix2Pix)" model variant enables instruction-based image editing. Rather than describing the desired output image, the prompt describes the editing operation to perform – the model understands the transformation as a directed action applied to the input image.

CORRECT USAGE

make it winter / convert to oil painting style / add dramatic storm clouds above

INCORRECT USAGE

snowy landscape with bare trees (this is a description, not an instruction)

Instruct Pix2Pix can be mixed with SD 1.5 base models for combined instruction-following and style guidance. Image guidance scale (separate from CFG) controls how strongly the input image structure is respected.

7.6 SCRIBBLE-TO-IMAGE

Draw Things provides a dedicated scribble canvas mode enabling rough hand-drawn compositional sketches to be used as img2img input, processed through ControlNet Scribble

conditioning. This enables rapid composition blocking without requiring a reference photograph or fully rendered input.

Workflow: Tap scribble canvas → draw rough shapes indicating major subject positions and scene elements → set prompt describing the intended scene → Draw Things routes the scribble through ControlNet Scribble → generates a coherent image respecting the compositional intent of the sketch.

SECTION 8 – INPAINTING & OUTPAINTING PIPELINE

8.1 INPAINTING PROTOCOL

Inpainting is the process of replacing a masked (erased) region of an existing image with AI-generated content. The model receives the unmasked regions as conditioning context and generates new content for the masked area that is semantically and stylistically consistent with the surrounding image.

Primary Use Cases:

- > Object removal (erase unwanted elements; fill with background-consistent content)
- > Object replacement (replace existing element with new subject)
- > Anatomy correction (fix extra fingers, distorted faces, limb errors)
- > Style-matching edits (add new elements that match the original image style)
- > Background extension (extend scene beyond original frame boundaries)

8.2 INPAINTING MODEL SELECTION

INPAINTING TOOLS

MODEL / TOOL	BASE ARCHITECTURE	BEST FOR	RESOLUTION	NOTES
FLUX.1 Fill	FLUX.1	Seamless backgrounds; object insertion; photorealistic	1024×1024	Best available inpaint quality; requires 16GB+ RAM
FLUX.2 (Kontext)	FLUX.2	In-context image editing; complex scene edits	1024×1024+	Newest; in-context editing without traditional masking in some modes
Foocus Inpaint LoRA	SDXL	Cartoon/detailed stylized edits	1024×1024	LoRA applied to SDXL base; strong for illustrated content
Alimama Inpainting ControlNet	FLUX.1	Complex multi-element scenes	1024×1024	ControlNet-guided inpainting; precise spatial control
Kwai Kolors Inpainting (8-bit)	SDXL-based	High-quality portrait and realistic edits	1024×1024	LoRA-compatible; excellent skin and clothing replacement
Wan 2.1 Fun Inpainting	Wan 2.1 video	Video inpainting; masked video frame replacement	480p	Fills masked regions in video frames consistently across temporal sequence

MODEL / TOOL	BASE ARCHITECTURE	BEST FOR	RESOLUTION	NOTES
Standard SD 1.5 Inpaint	SD 1.5	Simple edits; fast iteration; low-RAM devices	512×512	Fast; lower quality ceiling; widest device compatibility

8.3 INPAINTING OPERATIONAL PROCEDURE

- Step 1: Select the inpainting model from Model Manager (FLUX.1 Fill recommended if RAM permits).
- Step 2: Load or generate the base image on the Canvas.
- Step 3: Activate the Eraser tool. Mask only the target area – keep the mask boundary close to the object shape. Avoid masking large background areas unless necessary, as the model uses surrounding pixels as context.
- Step 4: Write the prompt describing ONLY the new content to generate in the masked area. Do not describe the surrounding image or the removal process. (Correct: red apple on wooden surface – Incorrect: replace the orange with a red apple)
- Step 5: Set Strength ≥70% for full replacement. Use 50–70% for texture/variation edits where structural preservation matters.
- Step 6: Set Mask Blur: 0–5 for sharp-edge content (illustration, cartoon, graphics). 10–30 for photorealistic content requiring seamless blending.
- Step 7: Generate. Review. Generate multiple seeds to select best match. Apply Server Offload for smaller device RAM management.

8.4 OUTPAINTING PROTOCOL

Outpainting extends the image canvas beyond its original borders, generating new content that seamlessly continues the existing scene. Draw Things supports an Infinite Canvas workflow for large panoramic or cinematic aspect ratio creation.

Procedure: Expand the canvas in the desired direction beyond the existing image boundary → write a prompt describing the new area to generate → use FLUX.1 Fill or equivalent fill model → generate. The model uses the existing image edge pixels as context for seamless extension.

For large panoramic creation: extend in increments of 50–70% of the original image width per outpainting pass. Overlapping the existing image in each pass ensures contextual continuity.

8.5 MASK BLUR CALIBRATION

BLUR VALUES BY CONTENT TYPE

CONTENT TYPE	RECOMMENDED BLUR VALUE	RATIONALE
Illustration / cartoon / line art	0-5	Sharp edges are stylistically appropriate; heavy feathering blurs line boundaries
Photorealistic (general)	10-20	Natural skin/texture transitions require gradual blending zone
Photorealistic (high-detail seamless)	20-30	Maximum feathering for invisible seams; use when boundary precision is critical
Background fill (object removal)	15-25	Background textures benefit from generous feathering to blend generated fill

8.6 COMMON INPAINTING FAILURE MODES & FIXES

FAILURE MODE RESOLUTION

FAILURE MODE	CAUSE	RESOLUTION
Model regenerates existing content unchanged	Prompt describes surrounding area rather than replacement content	Rewrite prompt to describe only the new target content
Style mismatch between inpainted region and original	Model style differs from source image style	Apply style-matching LoRA during inpaint; match model to original render style
Hard seam line visible at mask boundary	Mask blur too low	Increase mask blur to 15-25; re-generate
Inpainted region too conservative / low change	Strength <70% for full replacement tasks	Increase Strength to 75-90% for replacement; lower only for texture variation
Inpainted content leaks into unmasked areas	Mask too large; feathering extends too far	Reduce mask size; tighten mask to subject boundary
Anatomy errors persist through inpaint pass	Single-pass inpaint on complex anatomy	Multi-pass: inpaint hands separately from body; use face-specific inpaint pass

SECTION 9 – CONTROLNET & CONTROL SYSTEMS

9.1 CONTROLNET ARCHITECTURE

ControlNet is a system that extracts structural information from a reference image and uses that information to condition the generation process – effectively controlling composition, pose, edges, depth, and other spatial properties independently of the prompt. ControlNet is a *control* system rather than a *content* system: it constrains **what the generated image looks like in terms of structure**, while the prompt determines **what the generated image contains in terms of content**.

9.2 CONTROLNET TYPES & THEIR EFFECTS

CONTROLNET TYPE REFERENCE

CONTROLNET TYPE	WHAT IT EXTRACTS	BEST USE CASE	MODEL COMPATIBILITY	NOTES
Canny Edge	Edge lines and outlines	Preserving structure; line art; illustration conversion; manga/illustration generation	SD 1.5 / SDXL / SD3.5 (preview) / FLUX (custom)	Edge-preserving; adapts any image into "line drawing skeleton" to impose structure on new generation
OpenPose / Pose	Human skeletal keypoints	Character posing; figure composition; multi-person scenes	SD 1.5 / SDXL / FLUX	Extracts body pose from reference photo; enables character placement control
Depth / DPT	Spatial depth map	Landscape; perspective; scene composition; object placement in 3D space	SD 1.5 / SDXL / SD3.5	Generates grayscale depth representation; useful for maintaining spatial relationships
Scribble / Sketch	Hand-drawn sketches	Rapid composition blocking; converting sketches to finished art; scribble-to-image	SD 1.5 / SDXL / SD3.5	Custom sketch canvas in Draw Things; doodle a rough composition, generate realistic output
Normal Map	Surface orientation vectors	3D rendering; material surface detail; bump map-driven generation	SD 1.5 / SDXL	Extracts surface normals; useful for character/asset generation with consistent lighting
Semantic Segmentation	Object class masks	Complex multi-object composition; layout control; object count/position enforcement	SD 1.5 / SDXL	Assigns object categories to regions; advanced composition planning

CONTROLNET TYPE	WHAT IT EXTRACTS	BEST USE CASE	MODEL COMPATIBILITY	NOTES
IP-Adapter (Image Prompt)	Image content semantics	Style transfer; content referencing; "make it look like this"	SD 1.5 / SDXL	Injects semantic information from reference image; combines with ControlNet; model-specific versions
T2I Adapter (SDXL)	Multi-control unified conditioning	SDXL-specific multi-control; combining multiple spatial constraints	SDXL only	Lightweight; suitable for multi-control stacking; SDXL-optimized

9.3 CONTROLNET INSTALLATION & ACTIVATION

Draw Things v1.20250530.0+ supports the MiniSDXL suite for ControlNet.

Step 1 – Model Download: Navigate to Draw Things in-app ControlNet model library. Browse by type: Canny, OpenPose, Depth, Scribble, etc. Tap download on desired control models. Storage per model: ~10-30MB.

Step 2 – Activate Control Panel: In the generation panel, tap the ControlNet section to expand. Select the ControlNet type from the dropdown. Set the ControlNet strength scale (see Section 9.4 below).

Step 3 – Set Conditioning Input: Upload the control image (source pose photo, edge map, depth map, or sketch). The input image must match the ControlNet type.

Step 4 – Set Control Strength: Configure ControlNet weight/strength (see calibration table below).

Step 5 – Multi-Control Setup: Multiple ControlNets can be active simultaneously (e.g., OpenPose for body + Canny for background edges). Each has independent strength settings. Total control count: typically 2-3 simultaneous.

Step 6 – IP-Adapter Integration: Use image prompt alongside ControlNet for dual conditioning. Configurable as "Loaded from Image Prompt" in Draw Things v1.20250810.0+.

9.4 CONTROLNET STRENGTH CALIBRATION

STRENGTH VALUES BY TYPE

CONTROL TYPE	STRENGTH	EFFECT
Canny Edge	0.6-1.0	High strength preserves edge structure precisely; lower strength allows more interpretation
OpenPose / Pose	0.6-1.0	High strength locks body pose precisely; lower strength allows natural pose variation

CONTROL TYPE	STRENGTH	EFFECT
Depth / DPT	0.5-1.0	High strength locks spatial arrangement; lower allows compositional interpretation
Scribble / Sketch	0.8-1.0	High strength recommended; user sketches intentionally block rough composition
Normal Map	0.5-0.8	Moderate strength; surface orientation should guide but not dominate composition
Semantic Segmentation	0.4-0.8	Moderate strength; object layout guidance; too high restricts creative interpretation
IP-Adapter	0.4-0.8	Moderate strength; semantic transfer; too high causes style bleeding and dominant transfer
T2I Adapter (SDXL)	0.4-0.7	Lower strength recommended due to T2I's inherently strong conditioning behavior

FIELD TIP — MULTI-CONTROL STACKING

When combining multiple ControlNets, reduce individual strength by ~20% from single-ControlNet values. Example: OpenPose at 0.7 + Canny at 0.7 for a portrait with pose-locked figure but stylized background. Monitor for over-conditioning: if output looks "stiff" or "tight," reduce all ControlNet strengths by 0.1-0.2.

9.5 POSE CONTROL SPECIFICS

Draw Things provides a built-in Pose Editor with the following capabilities:

- > **Skeleton Detection:** Upload a reference photo; Draw Things automatically extracts the skeletal pose and maps it to the ControlNet conditioning
- > **Manual Joint Editing:** Individual joints can be repositioned on the skeleton to fine-tune the pose before conditioning generation
- > **Multi-Person Posing:** Supports up to 3 simultaneous skeleton detections and pose constraints in a single generation
- > **Pose Library:** Save frequently used poses for rapid reuse across generations without re-detection

SECTION 10 – VIDEO GENERATION PIPELINE

10.1 VIDEO GENERATION OVERVIEW

Draw Things supports both Text-to-Video (T2V) and Image-to-Video (I2V) generation natively on-device. The video pipeline uses diffusion model architecture extended with temporal attention layers that model relationships between consecutive frames. All processing runs locally on Apple Silicon with no cloud dependency.

10.2 VIDEO MODEL OPTIONS & HARDWARE REQUIREMENTS

VIDEO MODEL MATRIX

MODEL	PARAMS	TYPE	MAX RES	MAX FRAMES	MIN RAM	MOTION
Wan 2.1 (1.3B)	1.3B	T2V + I2V	480p	81 frames	8GB	Smooth; CausVid LoRA for 4-12 step acceleration
Wan 2.1 (14B)	14B	T2V + I2V	720p	81 frames	32GB	Best quality; strong motion coherence
LTX-2.3	2B	T2V + I2V	768×512	257+ frames	8GB	Real-time capable; very long sequences
Hunyuan Video	Variable	T2V + I2V	720p	Variable	16GB	Strong temporal coherence
ERNIE Image I2V	Variable	I2V	Variable	Variable	Variable	Newest pipeline (BAIDU, 2026)

10.3 TEXT-TO-VIDEO (T2V) PROTOCOL

Step 1 – Model Selection: Select Wan 2.1 or LTX-2.3 from Model Manager. Download the model (app handles installation automatically).

Step 2 – Prompt: Write a scene description. Video prompts benefit from specifying action, camera motion, and temporal flow rather than just a static scene description. "A camera slowly panning left across a tropical beach at sunset, waves gently rolling onto white sand"

Step 3 – Parameters:

- > Resolution: 480p (mobile) / 720p (desktop); higher = more RAM
- > Frame Count: 25-81 frames (shorter = faster, less RAM)
- > FPS: Typically 8 or 12 FPS; set via export settings post-generation
- > Sampler: Euler A Trailing (index 10) – universal default for all video models
- > Steps: 20-40 (quality increases above 25 with diminishing returns)
- > CFG: 3 for Wan 2.1; 1.5-5 range depending on prompt adherence vs. motion naturalness desired

Step 4 – Acceleration (Optional): Apply CausVid LoRA for Wan models. Set to 4-12 steps for dramatically faster generation. Quality lower but speed increase significant.

Step 5 – Generate: Tap Generate. Video diffusion takes 1-10 minutes depending on resolution, frame count, model size, and hardware.

10.4 IMAGE-TO-VIDEO (I2V) PROTOCOL

Step 1 – Image: Upload or generate the start frame on the Canvas.

Step 2 – Model: Select Wan 2.1 or LTX-2.3 I2V model variant.

Step 3 – Motion Description: Prompt describes desired motion from the still frame: "the camera slowly pushes forward; the character gently turns their head to the left; wind begins to rustle their hair."

Step 4 – Parameters: Same as T2V but with optional image conditioning strength. Higher image conditioning = more faithful to start frame. Lower = more creative motion but may deviate from original content.

Step 5 – Generate: The model extrapolates temporal evolution from the single-frame input.

10.5 CAUSVID LORA ACCELERATION

CausVid LoRA enables dramatically faster video generation through causal temporal compression – the model computes each frame sequentially building on previous frame predictions rather than computing all frames independently.

CAUSVID OPERATING PROTOCOL

PARAMETER	WAN 2.1 1.3B	WAN 2.1 14B
Causal Inference Setting	Disabled	Enabled

PARAMETER	WAN 2.1 1.3B	WAN 2.1 14B
CausVid Steps	4-12 (test range)	4-12
CausVid LoRA Model Source	In-app library or Hugging Face	In-app library or Hugging Face
Quality vs. Speed Tradeoff	4 steps = fastest, lowest quality; 12 steps = good balance	4 steps = viable draft; 12 steps = production-ready

10.6 VIDEO GENERATION BEST PRACTICES

- > Video generation requires significantly more RAM than image generation. On iPhone or iPad with <8GB RAM, enable Server Offload (see Section 11.5)
- > Shorter videos (25 frames, 480p) generate in 1-3 minutes on modern iPad
- > Motion coherence degrades beyond 49 frames for most models – break long video goals into segments
- > Use img2img + video sequence: generate high-quality still frames → run I2V on each → composite in post-production for long cinematic sequences
- > Video negative prompts have reduced effectiveness compared to image generation; focus on detailed positive motion descriptions instead
- > Frame rate selection: 8 FPS for stylized/artistic motion; 12 FPS for smoother natural motion; 15+ FPS via post-production frame interpolation

10.7 VIDEO EXPORT

Draw Things exports video in MOV format. Resolution options: 720p (desktop), 540p, 360p (mobile). FPS: configurable at export. Export directly to Photos app or Files for further editing.

SECTION 11 — ADVANCED WORKFLOWS & HYBRID OPTIMIZATION

11.1 SDXL REFINER PIPELINE

The SDXL Refiner is a specialized model trained to operate at the high-noise-to-low-noise transition point. It functions as a detail enhancement pass applied after the base SDXL model has generated the initial image structure.

SDXL Refiner Workflow:

1. Generate with SDXL Base: 1024×1024, 20-30 steps, DPM++ SDE Karras
2. Switch model to SDXL Refiner (v1.20250807.0+)  3. Set noise strength: 0.4-0.6 (40-60% noise retains ~50% original)
4. Generate refinement pass: 10-15 additional steps
5. Evaluate: may require 2-3 refinement passes depending on complexity

Multi-Refiner Strategy:

For maximum quality, apply the Refiner in multiple passes at progressively lower noise strengths: Pass 1 at 0.5 noise / 10 steps, Pass 2 at 0.35 noise / 8 steps, Pass 3 at 0.2 noise / 6 steps.

11.2 UPSCALING WORKFLOW

Draw Things supports native 2× upscaling within the generation pipeline.

Generate base image at native model resolution (512×512 for SD 1.5, 1024×1024 for SDXL)

Export the generated image

Re-import as img2img input at 2× target resolution (1024×1024 or 2048×2048)

Set Strength to 30-50%; write detailed, specific prompt

Generate: the diffusion process fills in high-frequency detail appropriate to the larger canvas size

Iterative 4× Upscaling:

For maximum resolution (e.g., 4096×4096): perform 2× upscale to 1024 → 2048, then a second 2× pass from 2048 → 4096. Between passes, evaluate and correct any artifacts that appear at the larger scale.

11.3 MODEL MIXING

Model mixing blends two checkpoints at a specified weight ratio, creating a hybrid model with characteristics of both parents.

Draw Things integration: Supported in-app via Model Manager → Model Mixing. Select two base checkpoints and blend ratio. Blended model saved as new checkpoint in library.

SPLIT MERGE (VERSION 1.20250818.0)

Split Merge enables partial layer merging by specified layer ranges rather than full-weight mixing. This preserves the structural qualities of the primary model while selectively blending specific visual style layers from the secondary model. Accessible via Model Manager → Split Merge with layer range input.

11.4 SERVER OFFLOAD MODE

Server Offload enables an iPhone or iPad to delegate generation to a local Mac running the gRPCServerCLI companion tool. The iOS device remains the primary interface (prompt entry, parameter control) while generation executes on the Mac's more powerful hardware.

Install gRPCServerCLI: Download from Draw Things GitHub. Run on Mac with the desired models installed locally.

Version Lock: gRPCServerCLI version must match Draw Things app version exactly. Update both in lockstep.

Connect: In Draw Things settings, enable Server Offload. The iOS device will auto-discover the Mac on the local network.

Specify Models: Configure which models are available on the server via the server's model configuration file.

Operation: Normal generation workflow; the generation is transparently routed to the Mac while the iOS device maintains full parameter control.

11.5 MULTI-PASS PRODUCTION WORKFLOW

For maximum quality production outputs:

Concept Pass: Generate initial composition with FLUX.1 [dev], Euler A Trailing, 1024×1024, 25 steps, CFG 2.5. Review composition and general concept.

Refinement Pass: Send to img2img at same resolution. Strength 50–70%. Refine prompt with specific detail additions. 25 steps.

Detail Correction Pass: Use Inpainting (FLUX.1 Fill). Fix anatomy errors, correct artifacts, refine specific areas. Mask blur 10–20. Strength 75–85%.

Upscale Pass: Export → img2img at 2× resolution. Strength 30–45%. Detailed prompt emphasizing texture and micro-detail. 25 steps.

Quality Enhancement (Optional): Send through SDXL Refiner (if using SDXL base) or apply LoRA style enhancement pass at low strength (0.3–0.5).

11.6 BATCH WORKFLOW

Batch processing generates multiple image variants from a single prompt configuration. Set a seed range or use "Generate All" to produce N variants with controlled diversity. Use fixed seed + prompt variations for systematic prompt A/B testing.

11.7 API SERVER

Draw Things includes a built-in API server (configured via Settings) providing REST endpoints compatible with the AUTOMATIC1111 API specification. Endpoints: /sdapi/v1/txt2img, /sdapi/v1/img2img, /sdapi/v1/options, /sdapi/v1/models.

API Server Use Cases: External script automation, workflow integration (e.g., Shortcut apps, custom scripts), batch processing via external tools, integration with third-party applications.

SECTION 12 – CONFIGURATION REFERENCE TABLES

12.1 QUICK-START SETTINGS BY USE CASE

READY-TO-DEPLOY PRESETS

USE CASE	MODEL	SAMPLER	STEPS	CFG	RESOLUTION	NEGATIVE
Photorealistic portrait	FLUX.1 [dev]	Euler A Trailing (10)	25	2.5	1024×1024	(minimal or omit)
Anime character	Illustrious SDXL	DPM++ SDE Karras (4)	25	6	1024×1024	worst quality, bad anatomy
Landscape / environment	FLUX.1 [dev]	Euler A Trailing (10)	28	3.0	1024×768	blurry, watermark
Fast concept sketch	SDXL Turbo	Euler A (1)	4	1	512×512	(none)
Logo / graphic design	FLUX.1 [schnell]	Euler A Trailing (10)	4	1	1024×1024	(none)
Vintage photograph	SD 1.5 + film LoRA	DPM++ 2M Karras (0)	25	7.5	512×512	digital art, 3D render, CGI
Text-in-image	Qwen Image 1.0	UniPC Trailing (17)	25	1.5	1024×1024	blurry, low quality, distorted text
Short mobile video	Wan 2.1 1.3B	Euler A Trailing (10)	20	3	480p, 25 frames	(none)
Cinematic video (desktop)	Wan 2.1 14B	Euler A Trailing (10)	30	3	720p, 49 frames	(none)
Image-to-video (desktop)	LTX-2.3	Euler A Trailing (10)	40	3	768×512	(none)

12.2 DEVICE CAPABILITY MATRIX

HARDWARE DEPLOYMENT LIMITS

DEVICE	RAM	MAX PRACTICAL MODEL	VIDEO CAPABLE	LORA TRAINING	NOTES
iPhone 12	4GB	SD 1.5 (8-bit)	No	No	Minimum supported; 8-bit only
iPhone 14 Pro	6GB	SDXL (8-bit)	Limited	No	Swap storage assists; SDXL viable
iPhone 15 Pro	8GB	FLUX.1 (quantized)	Wan 1.3B	Experimental	A17 Pro Neural Engine; 20% MFA 2.0 gain

DEVICE	RAM	MAX PRACTICAL MODEL	VIDEO CAPABLE	LORA TRAINING	NOTES
iPhone 16 Pro	8GB	Qwen Image (via swap)	Wan 1.3B	Limited	Advanced swap management
iPad (8GB RAM)	8GB	SDXL / FLUX quantized	Wan 1.3B	SD 1.5 LoRA	Min for LoRA training
M1 iPad Pro / M1 Mac	16GB	FLUX.1 [dev]	Wan 1.3B	SDXL LoRA	Comfortable FLUX usage
M2 / M3 Mac (18–36GB)	18–36GB	FLUX.1 [dev] / SD3.5	Wan 14B / Hunyuan	FLUX LoRA	20% faster on M3 (MFA 2.0)
M4 Pro Mac (24GB+)	24GB+	FLUX.1 [dev] 5-bit optimal	Wan 2.2 5B / Hunyuan	FLUX LoRA	Best inference per watt
M2 Ultra (192GB)	192GB	All models; no quantization	All video models	All LoRA types	Maximum tier; +25% vs mflux

12.3 SAMPLER COMPATIBILITY QUICK REFERENCE

COMPATIBLE SAMPLERS BY MODEL FAMILY

MODEL FAMILY	COMPATIBLE SAMPLERS	INCOMPATIBLE (AVOID)
SD 1.5	DPM++ 2M Karras (0), Euler A (1), DDIM (2), UniPC (3), DPM++ SDE Karras (4), LCM (6), Euler A Substep (7), DPM++ SDE Substep (8), TCD (9), DPM++ 2M AYS (12), Euler A AYS (13), DPM++ SDE AYS (14)	Trailing samplers (10, 11, 15, 16, 17); non-trailing samplers functional but not required
SDXL	DPM++ 2M Karras (0), DPM++ SDE Karras (4), UniPC (3), TCD (9), DPM++ 2M AYS (12), DPM++ SDE AYS (14), Euler A Substep (7)	Trailing samplers; LCM requires LCM-specific distilled model
FLUX.1 / FLUX.2	Euler A Trailing (10), DPM++ 2M Trailing (15), DDIM Trailing (16), UniPC Trailing (17)	ALL non-trailing samplers (0–9, 12–14)
SD3 / SD3.5	Euler A Trailing (10), DPM++ 2M Trailing (15), UniPC Trailing (17)	Non-trailing Karras and AYS samplers
Qwen Image	UniPC Trailing (17), Euler A Trailing (10)	Karras samplers; standard UniPC
Video (Wan/LTX/Hunyuan)	Euler A Trailing (10), DDIM Trailing (16), DPM++ SDE Trailing (11)	All non-trailing samplers

12.4 CFG SCALE BY MODEL FAMILY

RECOMMENDED CFG VALUES

MODEL FAMILY	RECOMMENDED CFG	ABSOLUTE MAX
SD 1.5 / SD 2.x	7–9	15
SDXL	5–7	10

SECTION 13 – TROUBLESHOOTING & FIELD NOTES

13.1 COMMON ISSUES & RESOLUTIONS

ISSUE RESOLUTION TABLE

SYMPTOM	CAUSE	RESOLUTION
CUDA-related errors	Incompatible hardware (NVidia GPU only)	Run on Apple Silicon or use CUDA-compatible backend; this is a hardware mismatch, not a configuration error
Install failed / model load error	Corrupted download; storage full	Delete model in Model Manager → redownload. Check free storage. Clear app cache if persistent
Import error on .safetensors	Non-standard weight format	Verify file is standard .safetensors. Try alternative source. Some community models use non-standard key naming
Model shows "Incompatible" warning	Sampler-model mismatch (v1.20250819.0+)	Switch to correct sampler (Trailing for FLUX/SD3; Karras for SD 1.5/SDXL)
VRAM error on generation	Model exceeds available RAM	Reduce resolution; reduce batch size; switch to 4-bit or 8-bit quantized model; use Server Offload; close other apps
Sampler not converging / different output each time	Ancestral sampler (Euler A, DPM++ SDE)	Ancestral samplers are stochastic by design. Use convergent sampler (DDIM, DPM++ 2M Karras) for identical results at fixed seed
Black image output	CFG too high; incompatible settings	Reduce CFG to 7 or below; ensure compatible sampler-model pairing
Grid / tiled output pattern	Resolution mismatch; VAE tiling	Ensure resolution matches model's native training resolution. Use multiples of 64. Disable VAE tiling
Over-saturated colors	CFG too high; over-conditioning	Reduce CFG. For SD 1.5/2.x: reduce below 10. For SDXL: reduce below 7. For FLUX: reduce below 3.5
Generated image ignoring prompt	CFG too low; insufficient prompt detail	Increase CFG (within model-safe range). Add descriptive detail to prompt. Check negative prompt is not overly restrictive
Bad anatomy (extra fingers, distorted faces)	Common SD 1.5 weakness	Use FLUX.1 [dev] or SDXL for anatomy-critical tasks. Add negative embeddings (<bad_hands_v5>). Use ControlNet OpenPose for pose control
LoRA not activating	Trigger word missing or incorrect	Verify exact trigger word from CivitAI model page. Check spelling and capitalization. Verify LoRA weight is above 0
LoRA output too weak / too strong	Weight mis-calibrated	Adjust weight: 0.5–0.75 for subtle, 0.8–1.0 for standard, 1.0–1.2 for strong. Reduce if artifacts appear

SYMPTOM	CAUSE	RESOLUTION
LoRA training crashes (iPhone)	Insufficient RAM for training	LoRA training is recommended on Mac with 16GB+. iPhone training is experimental. Reduce network dimension to 8-12
Inpainting leaves hard seam at mask boundary	Mask blur too low	Increase mask blur to 15-25. Use FLUX.1 Fill for best seam quality. Ensure prompt describes only replacement content
Inpainting regenerates unchanged content	Prompt describes surrounding area	Rewrite prompt to describe ONLY the new content to generate. Do not describe existing elements or removal
Video generation fails on iPhone	RAM insufficient	Reduce resolution to 480p; reduce frames to 25; use Wan 1.3B not 14B; enable Server Offload if Mac available
CausVid LoRA not accelerating	Settings incorrect	1.3B: disable Causal Inference; 14B: enable Causal Inference. Verify LoRA applied and weight = 1.0
Server Offload not connecting	Version mismatch; network issue	Ensure gRPCServerCLI version matches Draw Things app version. Both devices on same local network. Restart both services
API server connection refused	Firewall; wrong port; not enabled	Verify API server enabled in Settings. Check port configuration. Verify firewall allows connection. Default port: 7860

13.2 OPTIMIZATION CHECKLIST

// INFERENCE SPEED OPTIMIZATION

- > Reduce resolution: 512×512 generates faster than 1024×1024; 480p video faster than 720p
- > Reduce step count: 15–20 steps sufficient for many use cases; test before committing to 25+
- > Use distilled/small models: FLUX.1 [schnell] (4 steps) or SDXL Turbo (1–4 steps) for speed-first workflows
- > Quantize to 5–8-bit: Faster inference with minimal quality tradeoff. 4-bit if RAM is the binding constraint
- > Enable Metal FlashAttention v2 in Settings: ~20% faster inference on all Metal devices
- > Use convergent samplers: DPM++ 2M Karras (SD) or Euler A Trailing (FLUX) are fastest high-quality options
- > Batch related generations: Group generations that share model + resolution to minimize model reload overhead
- > Reduce model size: SD 1.5 (860M) generates significantly faster than FLUX.1 (12B); quality tradeoff acceptable for many tasks
- > Upgrade hardware: M3 is ~20% faster than M2 per MFA 2.0 gains; M4 ~20% faster than M3; M2 Ultra fastest available
- > Close background apps: Free device RAM reduces memory pressure and speeds up weight loading

13.3 PRIVACY & SECURITY

- > **All local generation:** When generating on-device with local models, zero data leaves the device. No internet connection is required during generation.
- > **Privacy Pass (v1.20250819.0):** For Draw Things+ cloud generation, Privacy Pass implements an unlinkable anonymization protocol. Cloud requests cannot be linked to specific user identities via zero-knowledge attestation.
- > **Model download:** Models downloaded from the in-app library or via CivitAI share only the download request metadata. No image data, prompt history, or generation data is transmitted.
- > **gRPC Server Offload:** Server Offload routes generation data over the local network only (Wi-Fi or LAN). No data traverses the internet when using Server Offload with a local Mac.
- > **API Server:** The built-in API server operates locally only. It does not expose endpoints to the public internet by default. Use with caution on untrusted networks.

> **Data minimization:** Draw Things does not collect or store generated images, prompts, or usage analytics unless the user explicitly enables cloud features via Draw Things+ subscription.

SECTION 14 – OUTLOOK & ROADMAP ANALYSIS

14.1 TRAJECTORY ASSESSMENT

Draw Things continues to evolve rapidly. The application maintains a consistent ~6-8 week major version cycle with each update delivering substantial new capabilities. The trajectory indicates continued investment in on-device performance, model diversity, and workflow efficiency.

14.2 SHORT-TERM OUTLOOK (NEXT 6 MONTHS)

- > ERDIE Image pipeline expansion: Additional BAIDU model integrations likely as ERNIE ecosystem matures
- > Video quality improvement: Wan 2.2 expected to advance motion coherence and temporal stability
- > Model manager enhancements: Continued expansion of in-app model library with one-click installs
- > API server improvements: Additional endpoints and tooling integration (v1.20251014.0 Bridge Mode foundation)
- > Hardware optimization: Continued Metal FlashAttention iteration; expect further 10-15% speed gains per major iOS/macOS release
- > Privacy Pass expansion: Zero-knowledge protocol likely extended to cover additional cloud features

14.3 MEDIUM-TERM OUTLOOK (6-18 MONTHS)

- > New model family support: Continued integration of emerging architectures (potential for Yi-image, additional BAIDU models, DeepSeek image models)
- > On-device training advancement: LoRA training RAM requirements likely to decrease through optimization; potential for full fine-tuning on high-end devices
- > Video generation scaling: Higher resolution video (1080p) likely on M3/M4-class hardware as models and optimizations advance
- > Cross-platform API standardization: Bridge Mode enabling deeper integration with external creative tools
- > Apple Silicon generation gains: M4 and subsequent generations expected to deliver continued per-watt efficiency improvements

14.4 LONG-TERM OUTLOOK (18+ MONTHS)

- > On-device training advancement: Continued reduction in RAM requirements for fine-tuning; potential for full-parameter fine-tuning on M2 Ultra-class hardware

- > Video generation scaling: Higher resolution, longer sequences, and improved temporal coherence through next-generation video diffusion architectures
- > Multi-modal integration: Continued convergence of image, video, and potentially audio generation capabilities within the same on-device deployment
- > API ecosystem maturity: Deep integration with creative industry workflows through comprehensive API support and third-party plugin ecosystems

14.5 KEY TAKEAWAYS

STRONGEST POSITION FLUX.1 [dev] on Apple Silicon delivers best-in-class image quality with full privacy, zero subscription, and offline capability. No competing Apple-native implementation matches this combination.	FASTEST GROWTH Video generation (Wan, LTX, Hunyuan) is the fastest-evolving capability. Expect significant quality and resolution improvements within 12 months.
MOST EFFECTIVE WORKFLOW Multi-pass generation (concept → refine → inpaint → upscale) produces production-grade results comparable to cloud-based pipelines.	KEY ADVANTAGE Draw Things' Metal FlashAttention 2.0 engine delivers 25% faster inference than competing implementations (mflux) and 94% faster than ggml/gguf on FLUX.1. This performance delta is widening.

III

MASTER MODEL INDEX

Model infrastructure assessment, calibration
parameters, and strategic deployment guidance
drawn from the full architectural index.

SECTION 15 – INFRASTRUCTURE: STRENGTHS & VULNERABILITIES

15.1 MODEL ARCHITECTURE SUMMARY

The Draw Things ecosystem supports three major architecture families: UNet-based models (SD 1.5, SD 2.x, SDXL and derivatives), DiT/Flow-based models (FLUX.1, FLUX.2, SD3/SD3.5, Wan video models), and LLM-based models (Qwen Image, HiDream, ERNIE Image). Each architecture has distinct operational characteristics that determine optimal deployment parameters.

15.2 UNET-BASED MODELS (SD 1.5 / SDXL ECOSYSTEM)

STRENGTHS	VULNERABILITIES
Widest LoRA/TI ecosystem (10,000+ community LoRAs on CivitAI). Mature, battle-tested. SDXL Turbo enables single-step generation. Fastest inference per step. Lowest RAM requirements. Broadest device compatibility (iPhone 12+).	Lower absolute quality ceiling vs. DiT models. Anatomy errors more common than FLUX/SD3. Text rendering weak. Limited to 77-154 tokens. Smaller parameter count (860M-3.5B) caps fine-detail capacity.

Deployment Guidance:

Use SD 1.5 for rapid iteration, device-constrained deployments, and tasks requiring massive LoRA ecosystem access. Use SDXL for balanced quality/speed on mid-range hardware. Avoid SD 1.5 for anatomy-critical, text-in-image, or maximum-quality production tasks where FLUX.1 is available.

15.3 DIT/FLOW-BASED MODELS (FLUX.1 / SD3 / VIDEO)

STRENGTHS	VULNERABILITIES
Highest quality ceiling. Superior text rendering. Multi-subject composition. 512+ token prompts. FLUX.1 [schnell] enables 4-step high-quality generation. Video temporal coherence. Scale-agnostic (FLUX.1 handles 1024-2048 natively).	Higher RAM requirements (16GB+ for full precision). Slower per-step inference. FLUX.1 [dev] non-commercial license. Negative prompt effectiveness reduced. Requires Trailing samplers. LoRA ecosystem smaller but growing rapidly.

Deployment Guidance:

Default to FLUX.1 [dev] for all production-quality still-image tasks on 16GB+ devices. Use FLUX.1 [schnell] for speed-first tasks where commercial licensing matters. Use FLUX.2 (Kontext) for in-context image editing. Use FLUX.1 Fill for inpainting. Quantize to 5-bit on M3/M4 devices for best RAM/quality balance.

15.4 LLM-BASED MODELS (QWEN IMAGE / ERNIE)

STRENGTHS	VULNERABILITIES
Best-in-class text rendering. Instruction-following capability. Multi-image referencing (Qwen). Extended context window. Natural language prompt processing without keyword formatting.	Highest RAM requirements (20GB+ for Qwen). Smaller LoRA ecosystem. Newer architecture with less community tooling. ERNIE Image is newest (2026) with evolving compatibility.

Deployment Guidance:

Use Qwen Image for text-in-image tasks where precise text rendering is the primary requirement. Use 4-step or 8-step acceleration LoRA to mitigate speed penalty. Avoid on devices with less than 20GB RAM. ERNIE Image should be considered experimental for production workflows until community testing confirms stability.

15.5 VIDEO MODEL INFRASTRUCTURE

VIDEO MODEL COMPARISON

ATTRIBUTE	WAN 2.1 1.3B	WAN 2.1 14B	LTX-2.3	HUNYUAN
Best for	Mobile video; fast iteration	Desktop quality; cinematic	Long sequences; real-time	Motion coherence; complex scenes
Max resolution	480p	720p	768x512	720p
Max frames	81	81	257+	Variable
Min RAM	8GB	32GB	8GB	16GB
CausVid support	Yes (disable Causal Inference)	Yes (enable Causal Inference)	N/A	N/A
Motion quality	Good	Excellent	Good (very long motion)	Excellent (temporal coherence)
Speed	Fast	Slow	Very fast	Moderate

15.6 SAMPLER INFRASTRUCTURE

The sampler ecosystem has bifurcated into two families:

- > **Karras/AYS Family (SD 1.5, SDXL):** DPM++ 2M Karras (0), DPM++ SDE Karras (4), UniPC (3), TCD (9). Optimized for traditional diffusion noise schedules. Do not use with flow models.
- > **Trailing Family (FLUX, SD3, video):** Euler A Trailing (10), DPM++ 2M Trailing (15), DDIM Trailing (16), UniPC Trailing (17). Designed for rectified flow matching timestep distribution. Required for flow-based models.

> **AYS Adaptive:** Concentrates compute at the most impactful timesteps. Enables 10-15 step outputs matching 25-30 step standard quality. Best for speed-optimized workflows on compatible models.

SECTION 16 — PROMPT CALIBRATION & EXECUTION PARAMETERS

16.1 PROMPT ARCHITECTURE BY MODEL

PROMPT STRATEGY

MODEL FAMILY	TOKEN LIMIT	OPTIMAL STYLE	WEIGHTING	QUALITY TAGS NEEDED
SD 1.5 / 2.x	77	Comma-separated keywords	Supported: (word:1.5)	Yes (highly effective)
SDXL	154	Natural language or keywords	Reduced sensitivity	No (unnecessary)
FLUX.1 / FLUX.2	512+	Long-form descriptive	NOT supported	No (minimal effect)
SD3 / SD3.5	512+	Natural language	Minimal support	No
Qwen Image	Extended	LLM-style instructions	Not applicable	No

16.2 CFG CALIBRATION PROTOCOL

Classical CFG saturation artifact onset is the primary risk above the recommended range. For flow models, the CFG effect is subtler and the effective range is compressed.

CFG CALIBRATION

MODEL FAMILY	RECOMMENDED	ABSOLUTE MAX	ARTIFACT ONSET	NOTES
SD 1.5 / 2.x	7-9	15	10+	Classical CFG; strong effect
SDXL	5-7	10	8+	Reduced vs. SD 1.5
FLUX.1 / FLUX.2	1-3.5	5	4+	Guidance distillation; compressed range
SD3 / SD3.5	4-7	10	8+	Flow model; moderate range
Qwen Image	1-2	3	2.5+	Very compressed range
Video models	1-5	7	5+	Motion quality degrades at high CFG

16.3 QUANTIZATION CALIBRATION

QUALITY VS. RAM TRADEOFF

QUANTIZATION	SIZE REDUCTION	QUALITY IMPACT	USE FOR
BF16 / FP16 (Full)	100%	None (reference)	M2 Ultra (192GB); maximum quality production

STRATEGIC DEPLOYMENT CONCLUSION

The Draw Things ecosystem represents the most capable on-device AI image and video generation platform available for Apple hardware. Its Metal FlashAttention 2.0 engine, broad model support (17+ architectures), and unique on-device LoRA training capability position it as the definitive tool for privacy-preserving, subscription-free creative AI.

DEPLOYMENT PRINCIPLES

1. Match the model to the hardware first – RAM is the binding constraint
2. Match the sampler to the model family – trailing for flow, Karras for SD
3. Calibrate CFG per model – SD 7-9, FLUX 1-3.5
4. Use negative embeddings for SD 1.5/SDXL anatomy
5. Omit negative prompts for FLUX/SD3 distilled models
6. Enable MFA 2.0 for 20% speed gain on all devices
7. Server Offload when RAM is the only barrier
8. CausVid for rapid video iteration; full steps for production
9. Multi-pass for production-grade stills
10. CivitAI integration for infinite LoRA/embedding access

HARDWARE RECOMMENDATIONS

Entry: iPhone 15 Pro – FLUX.1 (quantized), SDXL (8-bit)
Capable: M1 iPad Pro / M1 Mac 16GB – Full FLUX.1, SDXL LoRA training
Professional: M3 Pro 18GB+ – FLUX.1 (5-bit), FLUX LoRA training, video
Maximum: M2 Ultra 192GB – All models, all features, no quantization

STATUS – JANUARY 2026

Draw Things v1.20260501.0 is the current stable release. The application continues to receive regular updates with new model support and feature additions. The Metal FlashAttention 2.0 engine represents the state of the art for on-device diffusion inference on Apple Silicon. No competing implementation delivers comparable performance, model breadth, or feature depth within the Apple ecosystem.

IV

CLOUD FAILOVER PROTOCOL

Automated local AI infrastructure with
unrestricted model weight access, ComfyUI
workflow management, and persistent node-
graph deployment.

THE DEPLOYMENT STACK

INFRASTRUCTURE OVERVIEW

The Cloud Failover Protocol establishes a complete local AI infrastructure using Apple hardware. This protocol is designed to complement Draw Things' on-device capabilities with a local server-grade deployment, providing unrestricted model weight access, ComfyUI workflow management, and persistent node-graph deployment for advanced use cases.

STACK COMPONENTS

LAYER	COMPONENT	FUNCTION	ACCESS
Compute Layer	M2 Ultra Mac Studio (192GB RAM)	Primary inference engine; all models at full precision	Local network; SSH; VNC
Model Layer	Unrestricted .safetensors repository	Full community model access; no filtering; uncensored weights	Local filesystem; Hugging Face mirror
Workflow Layer	ComfyUI	Node-based pipeline construction; automation; multi-model workflows	Local web UI; API endpoints
Automation Layer	Python scripts + cron/systemd	Scheduled generation; batch processing; output pipeline	CLI; launchd/systemd
Storage Layer	APFS SSD + NAS	Persistent model storage; output archiving; backup	Local filesystem; SMB/NFS
Network Layer	gRPCServerCLI	iOS device offload; bridge between Draw Things mobile and Mac compute	Local network; gRPC protocol

ARCHITECTURE NOTE

This stack runs entirely on local infrastructure. No data leaves the local network. No cloud APIs are required. No subscription fees apply. All models are stored locally as unrestricted .safetensors weights with full community access.

INITIALIZATION SEQUENCE

STEP 1 – BASE SYSTEM

```
# macOS prerequisites
# Install Homebrew if not present
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

# Install Python 3.11 and dependencies
brew install python@3.11 git aria2

# Clone ComfyUI
git clone https://github.com/comfyanonymous/ComfyUI.git
cd ComfyUI
pip3 install -r requirements.txt

# Install Draw Things gRPCServerCLI
# Download from Draw Things GitHub releases page
# Place in /usr/local/bin/ and chmod +x
```

STEP 2 – GPU ACCELERATION

```
# For Apple Silicon (M1/M2/M3/M4)
# PyTorch uses Metal Performance Shaders (MPS) backend
# No additional driver installation required on macOS

# Verify MPS availability
python3 -c "import torch; print(torch.backends.mps.is_available())"

# Expected output: True (on Apple Silicon)
# If False: check macOS version (12.3+ required) and PyTorch version (2.0+)
```

STEP 3 – MODEL DIRECTORY STRUCTURE

```
ComfyUI/models/
  checkpoints/ # Base models (SD, SDXL, FLUX, etc.)
  loras/ # LoRA adapters
  vae/ # VAE models
  clip/ # CLIP text encoders
  controlnet/ # ControlNet models
  ipadapter/ # IP-Adapter models
  unet/ # Diffusion models (DiT/UNet)
  embeddings/ # Textual Inversion embeddings
  upscale_models/ # ESRGAN, Real-ESRGAN upscalers
```


PAYLOAD ACQUISITION – UNRESTRICTED WEIGHTS

MODEL SOURCES

WEIGHT ACQUISITION SOURCES

SOURCE	ACCESS	MODEL TYPES	NOTES
Hugging Face Hub	Direct download; git-lfs	All architectures	Primary source for official releases; community uploads; gated models require Hugging Face token
CivitAI	Web UI; API; Draw Things integration	SD 1.5, SDXL, FLUX LoRAs, embeddings	10,000+ community models; green play button integration with Draw Things
Black Forest Labs	Direct download; BFL API	FLUX.1 [dev], FLUX.1 [schnell], FLUX.2	Official FLUX weights; [dev] requires non-commercial license; [schnell] Apache 2.0
Stability AI	Direct download; Hugging Face	SDXL, SD3, SD3.5	Official Stable Diffusion weights; license terms vary by model
Draw Things In-App Library	One-tap download within app	Curated selection; verified compatibility	Convenience source; curated for known-good compatibility; not comprehensive
Local File Import	.safetensors via Files app / AirDrop	Any compatible .safetensors	Maximum flexibility; import any community model; verify format compatibility

DOWNLOAD PROTOCOL

```
# Method 1: Direct download with aria2 (recommended for large files)
aria2c -x 4 -s 4 "https://huggingface.co/black-forest-labs/FLUX.1-dev/resolve/main/flux1-dev.safetensors"

# Method 2: Hugging Face CLI
pip install huggingface-hub
huggingface-cli download black-forest-labs/FLUX.1-dev flux1-dev.safetensors --local-dir ./models

# Method 3: Git LFS
git lfs install
git clone https://huggingface.co/black-forest-labs/FLUX.1-dev

# Method 4: CivitAI with Draw Things integration
# Browse CivitAI in Safari → Tap green play button → Select "Draw Things"
# Draw Things auto-imports and begins download
```

// LEGAL NOTICE – UNRESTRICTED WEIGHTS

This protocol provides access to unrestricted model weights as published by their respective creators. Users are responsible for complying with individual model license terms. FLUX.1 [dev] carries a non-commercial license. FLUX.1 [schnell] is Apache 2.0. SDXL weights permit commercial use within Stability AI terms. Always verify license terms before commercial deployment. Draw Things, Inc. and this protocol do not endorse or facilitate circumvention of model license terms.

STORAGE & PERSISTENCE

STORAGE ARCHITECTURE

STORAGE LAYERS

LAYER	MEDIUM	CAPACITY	USE CASE
Primary (Hot)	M2 Ultra internal SSD	1-8TB	Active models; frequent access; inference loading
Secondary (Warm)	External Thunderbolt SSD	2-16TB	Model archive; less frequently used weights
Tertiary (Cold)	NAS (Network Attached Storage)	10-40TB	Long-term archive; backup; complete model repository
Mobile Cache	iOS/iPadOS device storage	128GB-2TB	Active models on mobile; cached weights for Server Offload

MODEL SIZE REFERENCE

TYPICAL STORAGE REQUIREMENTS

MODEL	FULL PRECISION	8-BIT	5-BIT	4-BIT
FLUX.1 [dev]	~22GB	~12GB	~8GB	~6GB
FLUX.1 [schnell]	~22GB	~12GB	~8GB	~6GB
SDXL Base	~7GB	~4GB	~3GB	~2GB
SD 1.5	~4GB	~2GB	~1.5GB	~1GB
SD3.5 Large	~16GB	~8GB	~6GB	~4GB
Wan 2.1 14B	~28GB	~15GB	~10GB	~8GB
Wan 2.1 1.3B	~3GB	~1.5GB	~1GB	~800MB

BACKUP PROTOCOL

```
# Automated backup script
# Add to crontab (daily at 3AM)
0 3 * * * rsync -av --progress /path/to/models/ /Volumes/NAS/ai-models-backup/

# Or use launchd on macOS
# Create plist in ~/Library/LaunchAgents/
# Load with: launchctl load ~/Library/LaunchAgents/com.user.ai-model-backup.plist
```

NODE GRAPH INJECTION

COMFYUI WORKFLOW PRINCIPLES

ComfyUI uses a node-graph paradigm where each generation pipeline is constructed as a directed acyclic graph of interconnected nodes. Nodes represent operations (model loading, sampling, VAE encoding/decoding, conditioning, upscaling) and edges represent data flow between operations.

The node-graph approach enables complex multi-model workflows that are not possible in single-panel interfaces: multi-model cascades, conditional branching, iterative refinement loops, parallel processing branches, and custom pipeline construction.

ESSENTIAL NODE TYPES

CORE NODES

NODE	FUNCTION	INPUTS	OUTPUTS
Load Checkpoint	Load base model weights	Checkpoint path	MODEL, CLIP, VAE
CLIP Text Encode	Encode prompt to conditioning	CLIP, text	CONDITIONING
KSampler	Execute diffusion denoising	MODEL, positive, negative, latent, seed, steps, cfg, sampler	LATENT
VAE Decode	Decode latent to pixel image	VAE, latent	IMAGE
Save Image	Save output to disk	IMAGE, filename_prefix	(none)
Load LoRA	Load and merge LoRA weights	MODEL, CLIP, lora_name, strength	MODEL, CLIP
Upscale Image	2x/4x upscale via model	IMAGE, upscale_model	IMAGE
Load ControlNet	Load ControlNet conditioning	ControlNet path, image	CONTROL_NET

SAMPLE WORKFLOW: TXT2IMG WITH LORA

```
[Load Checkpoint: SDXL Base] --(MODEL)--> [Load LoRA: Character LoRA]
--(MODEL)--> [KSampler]
--(CLIP)--> [CLIP Text Encode: Positive Prompt] --(CONDITIONING)--> [KSampler]
--(CLIP)--> [CLIP Text Encode: Negative Prompt] --(CONDITIONING)--> [KSampler]
--(VAE)--> [VAE Decode]
[Empty Latent Image: 1024x1024] --(LATENT)--> [KSampler] --(LATENT)--> [VAE Decode] --(IMAGE)-->
[Save Image]
[Load LoRA] --(CLIP)--> [CLIP Text Encode] (for trigger word injection)
```

SAMPLE WORKFLOW: MULTI-MODEL CASCADE

```
# Pass 1: Base generation with SDXL
[Load Checkpoint: SDXL] --(MODEL)--> [KSampler: 25 steps, cfg 6]
[Empty Latent: 1024x1024] --> [KSampler] --> [VAE Decode] --> [Save: base]

# Pass 2: Refinement with SDXL Refiner
[Load Checkpoint: SDXL Refiner] --(MODEL)--> [KSampler: 15 steps, cfg 5, denoise 0.4]
[VAE Encode: base image] --> [KSampler] --> [VAE Decode] --> [Save: refined]

# Pass 3: Upscale with Real-ESRGAN
[Load Upscale Model: Real-ESRGAN 4x+] --(MODEL)--> [Upscale Image]
[Save: refined] --(IMAGE)--> [Upscale Image] --(IMAGE)--> [Save: final 4096x4096]
```

WORKFLOW AUTOMATION

ComfyUI workflows can be saved as JSON files and loaded programmatically. Combine with cron/systemd for scheduled generation pipelines. Use the ComfyUI API server (launched with `--listen`) for external tool integration. API endpoint: `POST /prompt` with JSON workflow definition.

API INTEGRATION

```
# Start ComfyUI with API enabled
python3 main.py --listen 0.0.0.0 --port 8188

# Submit workflow via API
curl -X POST http://localhost:8188/prompt \\\n  -H "Content-Type: application/json" \\\br/>  -d
@workflow.json

# Monitor queue status
curl http://localhost:8188/queue

# Draw Things API server (built-in)
# Endpoints: /sdapi/v1/txt2img, /sdapi/v1/img2img, /sdapi/v1/options
# Enable in Draw Things: Settings → API Server → Enable
```

CONSOLIDATED REFERENCES

PRIMARY SOURCES

- > Draw Things App Store Listing (May 2026): <https://apps.apple.com/us/app/draw-things-ai-generation/id64444050820>
- > Draw Things Official Website: <https://drawthings.ai>
- > Draw Things GitHub Repository (gRPCServerCLI, model cards, documentation): <https://github.com/drawthingsai>
- > Metal FlashAttention 2.0 Engineering Blog (Draw Things): <https://liuliu.me/words/draw-things-metal-flashattention-2.0/>
- > Draw Things Subreddit (r/DrawThingsApp): Community troubleshooting, LoRA sharing, workflow discussion
- > Draw Things Discord Server: Official community; developer announcements; beta testing

MODEL SOURCES

- > Black Forest Labs (FLUX.1/FLUX.2): <https://huggingface.co/black-forest-labs>
- > Stability AI (SDXL, SD3, SD3.5): <https://huggingface.co/stabilityai>
- > Wan Video Models: <https://huggingface.co/Wan-AI>
- > Lightricks LTX-2.3: <https://github.com/Lightricks/LTX-Video>
- > Tencent Hunyuan Video: <https://huggingface.co/tencent/HunyuanVideo>
- > Alibaba Qwen Image: <https://huggingface.co/Qwen>
- > ERNIE Image (BAIDU): <https://huggingface.co/BAAI>
- > CivitAI Community Model Repository: <https://civitai.com> (10,000+ LoRAs, embeddings, checkpoints)
- > Hugging Face Model Hub: <https://huggingface.co/models>

TECHNICAL REFERENCES

- > FlashAttention (Dao et al.): <https://github.com/Dao-AILab/flash-attention>
- > Low-Rank Adaptation (LoRA) Paper: Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models" (ICLR 2022)
- > Textual Inversion Paper: Gal et al., "An Image is Worth One Word" (2022)
- > Classifier-Free Guidance Paper: Ho & Salimans (2021)
- > Karras et al. Diffusion Schedules: <https://arxiv.org/abs/2206.00364>

- > Stable Diffusion Paper: Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models" (CVPR 2022)
- > FLUX.1 Technical Report: Black Forest Labs (2024)
- > ComfyUI Documentation: <https://github.com/comfyanonymous/ComfyUI/wiki>
- > Apple Metal Performance Shaders Documentation: <https://developer.apple.com/metal/>

LICENSING & COMPLIANCE

- > FLUX.1 [dev]: Non-commercial license. Permits research, personal use, and evaluation. Commercial use requires separate license.
- > FLUX.1 [schnell]: Apache 2.0 license. Full commercial use permitted.
- > FLUX.2 (Kontext): License terms as specified by Black Forest Labs. Subject to change.
- > SDXL: Stability AI license. Permits commercial use within specified terms.
- > SD3 / SD3.5: Stability AI license. Terms subject to revision; verify current terms.
- > SD 1.5: CreativeML Open RAIL-M license. Permits commercial use with ethical use provisions.
- > Wan Video: Apache 2.0 license (verify per-model).
- > LTX-2.3: License as specified by Lightricks.
- > Community LoRAs/Embeddings: Individual creator licenses. Always verify per-file.

DOCUMENT METADATA & LEGAL

EDITION METADATA

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This document is provided for informational and educational purposes. All model names, trademarks, and registered trademarks are the property of their respective owners. Users are responsible for complying with all applicable software licenses, model license terms, and local regulations when deploying AI generation capabilities. The protocols described herein assume good faith use in accordance with applicable terms of service.

// CHANGE LOG – 2026.1

Added: ERNIE Image pipeline (BAIDU, 2026). Privacy Pass documentation. Bridge Mode API. CausVid LoRA expanded protocols. FLUX.2 (Kontext) full coverage. gRPCServerCLI version lock guidance. Split Merge (v1.20250818.0).

Updated: Device Capability Matrix for M4 Pro, iPhone 16 Pro. Sampler compatibility for 19-index system. Video model section expanded (Wan 2.2, LTX-2.3). Quantization recommendation matrix.

Revised: Troubleshooting section (21+ issue/resolution pairs). Ten deployment directives. CFG calibration ranges per model family. Model architecture assessment tables.

END OF DOCUMENT

DEPLOYMENT PROTOCOL //OMNIBUS EDITION 2026.1

Master Infrastructure Manual

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